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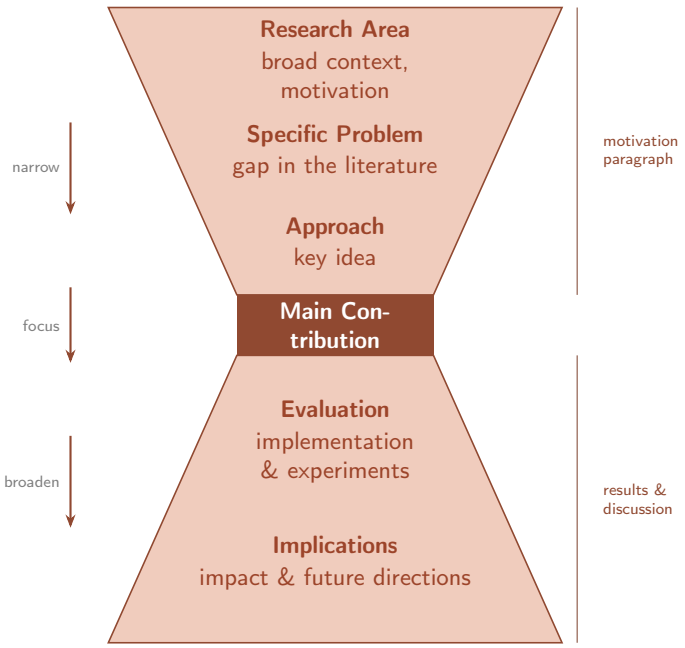
Expected Visiting Times in Interval Markov Chains

by
Jesper Nilsson

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First examiner:	Prof. Dr. Ir. Dr. h.c. Joost-Pieter Katoen
Second examiner:	apl. Prof. Dr. Thomas Noll
Thesis advisor:	Hannah Mertens

Abstract



Hourglass structure of a good abstract. Start broad (research area and problem), funnel down to the specific contribution, then widen again to state results and implications.

Write a self-contained summary of **roughly 150–250 words** following the hourglass structure above. Cover the following five points in order:

1. **Research area.** Name the field (e.g., strategy synthesis, probabilistic program verification, operational semantics) and state in one sentence what the field studies and why it matters.
2. **Specific problem.** State precisely which problem this thesis addresses and why it is challenging or not yet fully solved by existing approaches.
3. **Approach.** Describe the key idea of your contribution in one or two sentences. Avoid mathematical notation and abbreviations, and use easily understandable language.
4. **Theoretical results.** Name the main property proven (soundness, completeness, an equivalence result, or a complexity bound) and state it informally.

5. **Implementation and evaluation (if applicable).** Name the tool and the benchmark set and give the headline result: how many more examples are solved compared to the baseline and what is the runtime overhead.
6. **Implications.** Explain how your results solved the issue or filled the gap you pointed out beforehand.

Acknowledgements

I want to express my most profound appreciation to my supervisor!

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Chapter 1

Introduction

Stochastic processes are powerful Stochastic processes offer effective methods for analyzing real-world systems across fields like computer science, economics, and the social sciences [GT02, BLNW14, Afr20, Gal17]. One such widely used stochastic process are *discrete-time Markov chains* (DTMCs) which were introduced by Markov in 1906 [Mar06]. DTMCs consist of a set of states and a fixed transition probability function that specifies the probability of transitioning between states. They find application in a broad range of different domains, such as computer networks and software systems [GT02, PPH13], opinion dynamics [Gal17, FJ90], and biology [RC18].

When working with Markov Chains there exist multiple measures that are of interest, like reachability probabilities, stationary distributions, and expected rewards [Kat16]. Another quantity whose importance has been recognized by many authors [MKQW25, Cas09, RC18, MAK06, Afr20, GT02, SJT05, ST06, Gok04, MMAG11, PRT10, FRG13] are *expected visiting times* (EVTs), which measure how often a given model is expected to enter each state. Similarly to reachability probabilities and expected rewards, EVTs can be computed by finding the unique solution of a linear equation system [KS76].

These EVTs are used in various areas, such as determining the expected resource demands of software systems [GT02, GWHT04, SJT05, ST06] and provide insights for the quantitative evaluation of reliability [Gok04, MMAG11, PRT10] and security of computer systems [OA14, AN14].

However, a fixed probability function can not always be assumed in practice [JL91, Kuz91, Wal96, KU02], for example, when the probability function is estimated through statistical experiments, which only provide bounds on the transition probabilities. To capture this potential imprecision of the transition probability function, *interval-valued discrete-time Markov chains* (IDTMCs) were introduce in [JL91, KU02]. These IDTMCs extend DTMCs by defining a range within which the transition probability function is assumed to lie. Various semantics have been proposed to resolve the uncertainty which arises from this imprecision. Two widely used semantics are the

once-and-for-all semantic, also known as the *Uncertain Markov Chain (UMC) semantic*, [JL91, SVA06] and the *IMDP semantic* [SVA06].

As an IDTMC characterizes a set of DTMCs, the states of an IDTMC can have multiple different EVTs belonging to the contained DTMCs. We thus focus, in this thesis, on calculating the minimum and maximum EVTs for the states of an IDTMC. If we apply the aforementioned linear equation system, that can be used to calculate EVTs in DTMCs, to IDTMCs, it thus becomes an optimization problem. Moreover, the linear equation system depends on the specific transition probability function of the given DTMC. Therefore, when extending this linear equation system IDTMCs, we have to treat the transition probability function as variable as well, due to the imprecision of the transition probability function of IDTMCs. Thus, applying the linear equation system to IDTMCs to calculate the minimum and maximum EVTs necessitates solving nonlinear optimization problems. As many nonlinear optimization problems can only be solved in nondeterministic polynomial time (NP), we propose an alternative method to calculate the minimum and maximum EVTs for IDTMCs in this thesis. To our knowledge, this has not yet been studied in the literature.

1.1 Related Work

In this section, we elaborate on related work to position this thesis in the current research starting with the origins of EVTs. We continue by elaborating on topics such as expected rewards and reachability probabilities in IDTMCs.

The concept of EVTs was introduced in 1960 by Kemeny and Snell [KS76] for *absorbing DTMCs*, which consist of *absorbing* and *transient* states. The former describe states that cannot be left after they have been entered, i.e., these states only transition back to themselves. In contrast, the latter describe states that are eventually not entered again. The authors showed that the EVTs can be determined using the so-called *fundamental matrix*. In particular, the entry in the i-th row and j-th column of the fundamental matrix equals the EVT of the transient state s_j given that the unique initial state is the transient state s_i. Importantly, Kemeny and Snell also establish the notion of EVTs that coincide with the unique solution of a linear equation system.

As aforementioned, this thesis focuses on computing the minimum and maximum EVTs for the states of IDTMCs. There are multiple measures that are related to EVTs such as *expected rewards* [MKQW25] and *reachability probabilities* [BKL$^+$17, MKQW25]. Both of these measures have been studied in IDTMCs.

As shown in [CSH08] and [SVA06], IDTMC model checking under the IMDP semantic can be reduced to model checking a Markov Decision Process (MDP). There exist many algorithms and tools to calculate expected rewards for MDPs. However, this construction results in an "exponential blowup" as [Spr18] puts it, i.e., the constructed MDP is exponential in the size of the IDTMC [CSH08].

Moreover, as shown in [BDL+17], for every IDTMC under the UMC semantic, a parametric Markov Chain can be constructed that contains the same set of DTMCs as the corresponding IDTMC. Furthermore, expected rewards for pMCs with a fixed number of parameters is in PTIME [HBK17], i.e., computing expected reward for pMCs is exponential in the number of parameters [JJK22]. However, the construction shown in [BDL+17] creates a parameter for each interval. As we cannot bound the number of intervals of the IDTMCs without loss of generality this approach is not in PTIME.

Another approach

Nur kurz, wenn überhaupt

Furthermore, calculating minimum and maximum reachability probabilities in IDTMCs is in PTIME for both the UMC and IMDP semantic. Moreover, these two semantics coincide on reachability probabilities [BDL+17, CHK13]. To calculate the EVT of a state, we require two different reachabilty probabilities [BKL+17]. However, an optimal reachability probability is satisfied in an IDTMC if the reachability probability is fulfilled in at least one satisfying DTMC. When computing the minimum or maximum reachability probabilities for IDTMCs, we thus need to minimize or maximize these reachability probabilities

1.2 Contribution

1.3 Outline

Chapter 2

Preliminaries

In this chapter we introduce the theoretical foundations that are required and used throughout this thesis. We start in Sec. 2.1 by discussing the general notation used in this thesis. Rest of Overview of the sections ...

2.1 Notation

We denote the set of natural numbers, including zero, as $\mathbb{N}$ and the set of positive natural numbers, excluding zero, as $\mathbb{N}_{>0}$. The set of real numbers is denoted as $\mathbb{R}$ and the extended real numbers by $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ for which we use the following conventions:

(i) $x + \infty = \infty + x = \infty$ for all $x \in \mathbb{R}$,

(ii) $\infty + \infty = \infty$, and

(iii) $x \cdot \infty = \infty \cdot x = \begin{cases} \infty & \text{if } x \in \mathbb{R}_{>0}, \\ 0 & \text{if } x = 0, \\ -\infty & \text{if } x \in \mathbb{R}_{<0}. \end{cases}$

The conventions are defined analogously for $-\infty$. Other expressions such as $\infty - \infty$ and $\frac{\infty}{\infty}$ are undefined. For a given set A we denote its powerset as $2^X = \{Y \mid Y \subseteq X\}$.

Furthermore, we denote the entry in the i-th row and j-th column of a matrix $\mathcal{T}$, as $(\mathcal{T})_{ij}$. We denote the i-th entry of a vector v as $v(i)$. For convenience, we use the notation $\underline{n} = \{1, 2, \ldots, n\}$ with $n \in \mathbb{N}$. Moreover, for a given set S we use the notation $|S|$ to denote its cardinality.

Lastly, we use *Iverson Brackets* $[\![B]\!]$ to denote the function that maps a Boolean ex-

pression B to 1 or 0 such that

$$[\![B]\!] = \begin{cases} 1 & \text{if B is true,} \\ 0 & \text{otherwise.} \end{cases}$$

2.2 Theoretical Foundation

In this section, we introduce basic concepts from the subjects matrix, domain, measure, and probability theory which used throughout this thesis.

We start by introducing notions from domain theory. The provided definitions are taken from [MKQW25] whose material originates from [AGM95, Chapters 1].

Definition 2.1 (Partial Order)
Let X be a set with a relation $\sqsubseteq \subseteq X \times X$. The relation $\sqsubseteq$ is called a *partial order* if the following holds for all $x, y, z \in X$:

(i) $x \sqsubseteq x$ (reflexivity),

(ii) $x \sqsubseteq y$ and $y \sqsubseteq x$ implies $x = y$ (antisymmetry), and

(iii) $x \sqsubseteq y$ and $y \sqsubseteq z$ implies $x \sqsubseteq z$ (transitivity).

A partial order $\sqsubseteq$ on a set X is said to be a *linear order* if any two elements in X are comparable by $\sqsubseteq$.

We continue by briefly presenting some concepts from measure theory.

The following contents are taken from [MKQW25] whose material originates from [ADD00, Chapters 1, 4, and 5].

Definition 2.2 (σ-Algebra, Measurable Space)
Let Ω be a non-empty set. We call $\Sigma \subseteq 2^{\Omega}$ a *σ-algebra* on Ω if the following conditions are satisfied:

(i) $\Omega \in \Sigma$,

(ii) $X \in \Sigma$ implies $\Omega \backslash X \in \Sigma$ for all $X \subseteq \Omega$, and

(iii) $X_1, X_2, \cdots \in \Sigma$ implies $\bigcup_{i \in \mathbb{N}} X_i \in \Sigma$ for all $X_1, X_2, \cdots \subseteq \Omega$.

If Σ is a *sigma*-algebra on Ω, (Ω, Σ) is called a *measurable space*.

For the extended real numbers $\overline{\mathbb{R}}$, we consider the measurable space $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$, where $\mathcal{B}(\overline{\mathbb{R}})$ is the *Borel σ-algebra* on the extended real numbers.

Definition 2.3 (Borel σ-Algebra on the Extended Real Numbers)
The Borel σ-algebra on $\overline{\mathbb{R}}$ is defined as the smallest σ-algebra $\mathcal{B}(\overline{\mathbb{R}})$ containing all intervals $(a, b] = \{x \in \overline{\mathbb{R}} \mid a < x \leq b\}$ with $a, b \in \overline{\mathbb{R}}$.

For a measurable space (Ω, Σ), a set $A \subseteq \Omega$ is called *measurable* if $A \in \Sigma$. The measurability of a function is defined as follows.

Definition 2.4 ((Borel) Measurable Function)
Let (Ω, Σ) and (Ω', Σ') be measurable spaces. A function $f : \Omega \to \Omega'$ is called *measurable* relative to Σ and Σ' if $\{\omega \in \Omega \mid f(\omega) \in f(E)\} \in \Omega$ for all $E \in \Sigma'$.

A function $f : \Omega \to \overline{\mathbb{R}}$ is called *Borel measurable* (on (Ω, Σ)) if it is measurable relative to Σ and $\mathcal{B}(\overline{\mathbb{R}})$.

Moreover, the class of Borel measurable functions is closed under addition.

Theorem 2.5 (Addition of Borel Measurable Functions)
Let (Ω, Σ) be a measurable space and let $f, g : \Omega \to \overline{\mathbb{R}}$ be two Borel measurable functions. The function $f + g$, defined by pointwise addition, is measurable if it is well-defined, i.e., if $f(\omega) = \infty$ implies $g(\omega) \neq -\infty$ and $f(\omega) = -\infty$ implies $g(\omega) \neq \infty$ for all $\omega \in \Omega$.

Additionally, to discuss the probability of measurable sets, we introduce the notion of *probability measures*.

Definition 2.6 (Probability Measure, Probability Space)
Let (Ω, Σ) be a measurable space. A function $\Pr : \Sigma \to [0, 1]$ is a *probability measure* on (Ω, Σ) if $\Pr(\Omega) = 1$ and for every countable collection $A_1, A_2, \ldots \in \Sigma$ of disjoint sets it holds that

$$\Pr\left(\bigcup_{k \geq 1} A_k\right) = \sum_{k \geq 1} \Pr(A_k).$$

If $\Pr : \Sigma \to [0, 1]$ is a probability measure on (Ω, Σ), then $(\Omega, \Sigma, \Pr)$ is called a *probability space*.

In a probability space $(\Omega, \Sigma, \Pr)$, the elements of Ω are called outcomes, while the measurable subsets of Ω are referred to as events. An event $E \in \Sigma$ is said to hold *almost surely* if $\Pr(E) = 1$.

In this thesis, we make use of the Lebesgue integral $\int_\Omega f(\omega) dPr(\omega)$ of a Borel measurable function $f : \Omega \to \overline{\mathbb{R}}$ and a probabiltiy measure Pr on a probability space (Ω, Σ). We presuppose that the reader is familiar with the notion of Lebesgue integration. A thorough examination can be found in [ADD00, Chapter 1.5].

Definition 2.7 (Expected Values)
Let (Ω, Σ, Pr) be a probability space. If $f : \Omega \to \overline{\mathbb{R}}$ is a Borel measurable function on (Ω, Σ, Pr), then the *expected value* of f, denoted as $\mathbb{E}[f]$, is the Lebesgue integral of f with respect to the probability measure Pr, i.e., $\mathbb{E}[f] = \int_\Omega f(\omega) dPr(\omega)$, provided the integral is well-defined.

A *probability distribution* is a function that assigns probabilities to outcomes.

Definition 2.8 (Discrete Probability Distribution)
A function P is a *discrete probability distribution* with $P : X \to [0, 1]$, where X is a countable set and $\sum_{x \in X} P(x) = 1$.

We continue by introducing notions from matrix theory based on [**?**].

An important notion, which we require in this thesis, are *(strictly) row-stochastic matrices*.

Definition 2.9 (Row-Stochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-stochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$.

When considering each row as a function $f_i(j) = (\mathcal{T})_{ij}$, these functions are discrete probability distributions.

Definition 2.10 ((Strictly) Row-Substochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-substochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} \leq 1$ for all rows $i \in \underline{n}$.
$\mathcal{T}$ is strictly row-substochastic if and only if ther exists at least one row i' with $\sum_{j=0}^{n} (\mathcal{T})_{i'j} < 1$

If a matrix A converges to the null matrix, then the inverse of $(\boldsymbol{I} - A)$, where $\boldsymbol{I}$ is the identity matrix, is equal to the so-called Neumann series [Neu77].

Definition 2.11 (Matrix inversion by the Neumann Series)
Given a matrix $A \in \mathbb{R}^{n \times n}$ with $\lim_{k \to \infty} A^k = 0$, i.e. A is convergent, the Neumann series $\sum_{k=0}^{\infty} A^k$ converges against $(\boldsymbol{I} - A)^{-1}$.

We will now shortly introduce linear and nonlinear programming based on [LY08].

Definition 2.12 (Linear Programming)
A *linear programming* problem with n variables and m constraints can be formu-

lated as:

$$\begin{aligned} \text{minimize} \quad & c^\mathsf{T} x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \ , \end{aligned}$$

where $c, x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$.

In contrast to a linear programming, a nonlinear programming has a nonlinear objective function or nonlinear constraints.

Definition 2.13 (Nonlinear Programming)
A *nonlinear programming* problem with n variables and $m + r$ constraints can be formulated as:

$$\begin{aligned} \text{minimize} \quad & f(x) & & \\ \text{s.t.} \quad & g_i(x) \leq 0 & & i \in \underline{m} \\ & h_j(x) = 0 & & j \in \underline{r} \\ & x \in S \ , & & \end{aligned}$$

where $x \in \mathbb{R}^n$ and $f, g_i, h_j : S^n \to \mathbb{R}$ for all $i \in \underline{m}$, $j \in \underline{r}$. At least one of the functions f, g_i, or h_j is nonlinear.

Lastly, we use the convergence of the geometric series in this thesis.

Definition 2.14 (Convergence of Geometric Series)
The geometric series $\sum_{k=0}^{\infty} q^k$ converges for all $q \in \mathbb{R}$ with $|q| < 1$, where

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \ , \quad \text{for } |q| < 1$$

Chapter 3

Markov Chains

The previous chapter provided preliminary mathematical notions, including .... In the following we introduce the notion of Markov Chains, which we require in (the main part of this) thesis.

This chapter starts with a formal definition of Markov Chains in Sec. 3.1. We then continue, by elaborating on related concepts, such as the probability space, reachability probabilities, as well as recurrent and transient states. Furthermore, we describe a further important concept, the *expected visiting times* of states in DTMCs, in Sec. 3.2 which were first introduced by [KS76].

Lastly, we relate these *expected visiting times* to reachability probabilities in Sec. 3.3. (For which we provide a short proof ...)

3.1 Discrete-Time Markov Chains

In this section, we will introduce definitions and theorems for Markov Chains. The notation and contents of this section are mostly based on [BK08], [MKQW25], and [CSH08].

Definition 3.1 (DTMC)
A discrete-time Markov chain (DTMC) is a tuple $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota_{init}^{\mathcal{D}})$, where

(i) S is a countable, nonempty set of states,

(ii) $\mathbf{P} : S \times S \to [0,1]$ is the *transition probability function*, with $\sum_{s' \in S} \mathbf{P}(s,s') = 1$ for all states $s \in S$, and

(iii) $\iota_{init} : S \to [0,1]$ is the *initial distribution*, such that $\sum_{s \in S} \iota_{init}(s) = 1$.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $S^{\mathcal{D}}$, $\mathbf{P}^{\mathcal{D}}$, and $\iota_{init}^{\mathcal{D}}$. We assume in this thesis, that the DTMCs are *finite*, i.e., that the

set of states S is *finite*. We call a state $s \in S$ is called *initial* if it satisfies $\iota_{init}(s) > 0$. Furthermore, we assume an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s_i, s_j \in S$ and $i < j$. This allows us, for example, to denote the i-th entry of the initial distribution as $\iota_{init}(s_i)$, when viewing ι_{init} as a column vector. Due to the conditions (i) and (ii) provided in Def. 3.1, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions, see Def. 2.8.

Moreover, if the probability to transition from a state s_i to s_j, for two given states $s_i, s_j \in S$, is greater than zero, i.e., $\mathbf{P}(s_i, s_j) > 0$, then we refer to s_j as a direct successor of s_i. Conversely, we refer to s_i as a direct predecessor of s_j. Additionally, we define a neighbour set N, with $s_j \in N(s_i)$, if and only if s_j is a direct successor of s_i.

Paths and Probability Space. To reason about the probabilistic behavior of a DTMC $\mathcal{D}$, we define its probability space, in which the outcomes correspond to the infinite paths of $\mathcal{D}$.

To this end, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ for the rest of this section.

Definition 3.2 (Path of a DTMC)
An *infinite path* π in a DTMC $\mathcal{D}$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S$, such that $\mathbf{P}(\pi[k], \pi[k+1]) > 0$. A *finite path* is a finite prefix of an *infinite path*.

We use the notation $\pi[k] \in S$ to denote the k-th element of the path π for all $k \in \mathbb{N}_0$, i.e. $\pi[k] = s_k$.

Notice, that for any path π a given state $\pi[k]$, which is visited in step $k \in \mathbb{N}$, is always followed by one of its direct succesors, i.e., $\pi[k+1] \in N(\pi[k])$. Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \dots, s_k) \in Paths^{\mathcal{D}}_{fin}$ to be equal to k, which we denote as $|\pi| = k$. As an infinite path is infinite in lenght, we define $\pi \in Paths^{\mathcal{D}}_{inf}$ to be $|\pi| = \infty$. We denote the last state of a finite path $\pi \in Paths^{\mathcal{I}}_{fin}$ as $last(\pi)$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$. The sets of all infinite and finite sets are denoted as $Paths^{\mathcal{D}}_{inf}$ and $Paths^{\mathcal{D}}_{fin}$, respectively.

Further, we construct the *probability space* of a DTMC as follows. The events correspond to sets of infinite paths that can be expressed as countable unions and intersections of cylinder sets, i.e., sets containing all possible continuations of finite paths in $\mathcal{D}$.

Definition 3.3 (Probability Space of a DTMC)
The *probability space* of a DTMC $\mathcal{D}$ is defined as $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}}, Pr^{\mathcal{D}})$, where

- $\sum_{Paths^{\mathcal{D}}_{inf}}$ denotes the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$

for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$ with

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\}$$

- $Pr^{\mathcal{D}}$ is the unique probability measure on $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}})$ satisfying

$$Pr^{\mathcal{D}}(Cyl(\pi')) = \iota^{\mathcal{D}}_{init}(\pi'[0]) \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{D}}(\pi'[k], \pi'[k+1])$$

for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$

Additionally, we use the notation $Pr^{\mathcal{D}}_s$ to denote the probability measure on the measurable space induced by fixing $s \in S$ as the unique initial state. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $Pr^{\mathcal{D}}$ and $Pr^{\mathcal{D}}_s$.

Intermediate States. Using the notion of finite paths, we introduce *intermediate states*, which we later require in Sec. 4.1.1.

Definition 3.4 (Intermediate States)
Given a DTMC $\mathcal{D}$, a set of states T, and a state $s \in S$, then $u \in S$ is *intermediate* if and only if there exists path $\pi \in Paths^{\mathcal{D}}_{fin}$ with $|\pi| = n$, such that $\pi[0] \in T$, $\pi[n] = s$, and $\pi[k] = u$ for some $k \in \underline{n}$.

In other words, there exists a finite path from some $t \in T$ to s in $\mathcal{D}$ that visits u. We denote the set of these intermediate states as $I^{\mathcal{D}}_{T,s}$ for a set of states $T \in S$ and state $s \in S$. Moreover, if T is a singleton set, i.e., $T = \{t\}$, then we simply denote the set of intermediate states as $I_{t,s}$.

Reachability probability. In the following, we consider events, i.e., measurable sets of infinite paths, which are relevant for the introduction of reachability probabilities in DTMCs.

First, we define the *until* operator (U), see [BK08].

Definition 3.5 (Until Operator)
Let $R, T \subseteq S^{\mathcal{D}}$ be two sets of states of a DTMC $\mathcal{D}$. For these two sets, we define the *until* operator as

$$R \,\mathsf{U}\, T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Futhermore, the *step-bounded until* operator is defined as

$$R \,\mathsf{U}^{=k}\, T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Intuitively, the set $R \mathsf{U} T$ contains all paths that only visit states in R, until a state in T is visited.

Using the *until* operator, we can introduce the *eventually* operator ($\Diamond$).

Definition 3.6 (Eventually Operator)
Let $T \subseteq S$ be a set of states of a DTMC $\mathcal{D}$. Then we define the *eventually* operator as $\Diamond T = S \mathsf{U} T$.

Consequently, we define the *step-bounded eventually* operator as $\Diamond^{=k} T = S \mathsf{U}^{=k} T$.

In other words, $\Diamond T$ contains all paths $\pi \in Paths^{\mathcal{D}}_{fin}$ that visit at least one $t \in T$ at some point, i.e., there exists at least one $k \in \mathbb{N}$ with $k \leq |\pi|$, such that $\pi[k] = t$.

With the *eventually* operator, we can define the *reachability probability*.

Definition 3.7 (Reachability Probability)
The reachability probability of a set $T \subseteq S^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is given by $Pr(\Diamond T)$.

Thus, the reachability probability describes the probability of visiting a set of states in a DTMC $\mathcal{D}$. Intuitively, if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable in $\mathcal{D}$, i.e., there exists at least one $\pi \in Paths^{\mathcal{D}}_{fin}$, such that $\pi[0]$ is an initial state of $\mathcal{D}$ and π visits t. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$ (and $S \cup t$ instead of $S \cup T$). Lastly, we denote $\Diamond_t T = \{\pi \in \Diamond T \mid \pi[0] = t\}$.

To calculate the reachability probability in this thesis, we often use the recursive formula, found in the following lemma.

Lemma 3.8 (Recursive Formula for Reachability Probability)
For states $t \in S$ and $T \subseteq S$:

$$Pr_t(\Diamond T) = \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}(t,u) \cdot Pr_u(\Diamond T)$$

with $Pr_t(\Diamond T) = 1$ for all $t \in T$.

We will provide a short proof for this lemma.

Proof 1
Using Def. 3.3,

$$
\begin{aligned}
Pr_t^{\mathcal{D}}(\Diamond T) &= \int_{\pi \in \Diamond T} \iota_{init}^{\mathcal{D}}(\pi[0]) \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \int_{\pi \in \Diamond_t T} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as t is the unique initial state, i.e., } \iota_{init}(t) = 1.\text{)} \\
&= \sum_{u \in N(t)} \int_{\pi \in \Diamond_t T \wedge \pi[1]=u} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as } t \text{ can only be succeded by } u \in N(t)\text{)} \\
&= \sum_{u \in N(t)} \mathbf{P}(t,u) \int_{\pi \in \Diamond_t T \wedge \pi[1]=u} \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(with } \pi[0] = t, \pi[1] = u\text{)}
\end{aligned}
$$

For $\pi \in \Diamond_t T$ with $\pi[1] = u$, define $\pi' = (\pi[1], \dots, last(\pi))$. As $\pi'[0] = u$ and $last(\pi') = s$, $\pi' \in \Diamond_u T$. Conversely, for $\pi' \in \Diamond_u T$, define $\pi = (t, \pi'[0], \dots, last(\pi'))$, then $\pi \in \Diamond_t T$. Thus,

$$
\begin{aligned}
&= \sum_{u \in N(t)} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) \\
&= \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) && \text{(as, if } u \notin I_{tT}\text{, then } Pr_u(\Diamond T) = 0\text{, see Def. 3.4.)} \\
& && (3.1)
\end{aligned}
$$

□

Lastly, we use the *next* operator ($\bigcirc$), which is defined as follows.

Definition 3.9 (Next Operator)
Let $\mathcal{D}$ be a DTMC and let $E \in \sum_{Paths_{inf}^{\mathcal{D}}}$ be a measurable set of infinite paths in $\mathcal{D}$. We now define the *next* operator, as

$$\bigcirc E = \{\pi \in Paths_{inf}^{\mathcal{D}} \mid (\pi[1], \pi[2], \dots) \in E\} ,$$

In other words, a path $\pi \in Paths_{inf}^{\mathcal{D}}$ is contained in $\bigcirc E$ if and only if $\pi' = (\pi[1], \pi[2], \dots)$ is contained in E. Moreover, the next step probability is defined as $Pr(\bigcirc E) = \sum_{s' \in N(s)} \mathbf{P}(s,s') \cdot Pr_{s'}(E)$.

In this thesis, we use the *next* operator ($\bigcirc$) in combination with the *eventually* operator ($\Diamond$). Specifically, we use them to denote the *recurrence probability* $Pr_s(\bigcirc\Diamond s)$, i.e., the probability to reach s from all of its successors.

Strongly Connected Components. We can partition the state set of a DTMC as shown in the following definition.

Definition 3.10 ((Bottom) Strongly Connected Component)
A non-empty set of states $C \subseteq S^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is *strongly connected* if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths^{\mathcal{D}}_{inf}$, such that $\pi[0] = s$ and $\pi[k]$ for some $k \in \mathbb{N}$. If C is strongly connected and no proper subset of C, i.e. $C' \subset C$, is strongly connected, then C is called a *strongly connected component* (SCC).

Furthermore, an SCC C is called a *bottom strongly connected component* (BSCC) if there exists no $k \in \mathbb{N}$ and no path $(\pi[0], \dots, \pi[k], \dots) \in Paths^{\mathcal{D}}_{inf}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.

In other words, once a state of a BSCC C is entered, C cannot be left again.

Using BSCCs, we can define *acyclic* DTMCs.

Definition 3.11 (Acyclic DTMC)
A DTMC $\mathcal{D}$ is called *acyclic* if and only if each SCC is either a BSSC or a singleton set $\{s\}$ with $\mathbf{P}(s, s) = 0$

Recurrent and Transient States. Using the *recurrence probability* $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s)$, we can partition the state set into *recurrent* and *transient* states.

Definition 3.12 (Recurrent and Transient States)
For a given DTMC $\mathcal{D}$ and state $s \in S$, s is called *recurrent* if and only if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not *recurrent* are called *transient*.

In other words, a state is *recurrent* precisely, when we are guaranteed to reach it from all of its direct successors, meaning once we reach a *recurrent* state, we are bound to return to it. Notice, that *recurrent* states are precisely the states contained in a BSCC. Using this definition, we can split the states S of a DTMC into two sets with S_{tr} containing all of the transient states and S_{re} containing all of the recurrent states. For convenience, we additionally define the set of *initial* states $S^{\mathcal{D}}_{in} = \{s \mid \iota^{\mathcal{D}}_{init}(s) > 0\}$. Furthermore, we define the set of *transient initial* states $S_{tr \cap in} \subseteq S_{in}$ with $S_{tr \cap in} = S_{in} \cap S_{tr}$.

Moreover, the event $\Diamond S_{re}$, i.e., to reach any recurrent state, holds *almost surely* in any DTMC.

Theorem 3.13 (Limit Behaviour)
For any DTMC $\mathcal{D}$, set of recurrent states S_{re}, and any state $s \in S$, it holds that $\lim_{k \to \infty} Pr_s(\Diamond^{=k} S_{re}) = 1$.

Intuitively, the probability of reaching a recurrent state after k steps, starting in s,

tends towards 1 as k tends towards infinity.

Lastly, we define *absorbing* states.

Definition 3.14 (Absorbing state)
A states s of a DTMC $\mathcal{D}$ is refered to as an *absorbing state* if and only if $\mathbf{P}^{\mathcal{D}}(s,s) = 1$.

This means, once the model enters an *absorbing* state, it cannot leave it. Notice, that *absorbing* states are a special type of *recurrent* states, as the only direct successor of an *absorbing* state $s \in S$ is s itself and therefore trivially $Pr_s(\Diamond s) = 1$. Thus, we are guaranteed to return to a given *absorbing* state after one step.

Definition 3.15 (Markov Descision Process (MDP))
[BK08] A *Markov Decision Process* (MDP) is a tuple $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$, where

(i) S is a countable set of states

(ii) Act is a set of actions

(iii) $A(s) \subseteq Act$ denotes the applicable actions for all states $s \in S$

(iv) $\mathbf{P} : S \times Act \times S \to [0,1]$ is the transition probability function such that $\sum_{s' \in S} \mathbf{P}(s, \alpha, s') = 1$ for all $s \in S$ and $\alpha \in A(s)$

(v) $\iota_{init} : S \to [0,1]$ is the initial distribution such that $\sum_{s \in S} \iota_{init}(s) = 1$

Definition 3.16 (Canonical Form)
For a DTMC $\mathcal{D}$, its transition matrix $\mathcal{T}$ is said to be in canonical form if and only if

$$\mathcal{T} = \begin{bmatrix} Q & R \\ 0 & S \end{bmatrix}$$

where

(i) $Q = (\mathbf{P}(s,t))_{s,t \in S_{tr}}$ contains the transition probabilities between the transient states

(ii) $R = (\mathbf{P}(s,t))_{s \in S_{tr}, t \in S_{re}}$ contains the transition probabilities between transient and recurrent state pairs

(iii) $S = (\mathbf{P}(s,t))_{s,t \in S_{re}}$ contains the transition probabilities between the recurrent states

For absorbing DTMCs $S = \boldsymbol{I}$. As mentioned in **??**, this is precisely possible for

3.2 Expected Visiting Times for DTMCs

In this section, we will introduce the concept of *expected visiting times*, which were first introduced by Kemeny and Snell [KS76] in the context of the *fundamental matrix*. The notation and contents of this section are mostly based on [MKQW25] and [KS76].

Intuitively, the notion of EVTs describes how often we expect the model to be in a given state. To formally introduce EVTs, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ and start by introducing *visiting times* along paths.

Definition 3.17 (Visiting Times of Infinite Paths)
The function $vt_s : Paths^{\mathcal{D}}_{inf} \to \mathbb{R}$ describes the *visiting time* of a state $s \in S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!]$$

As proven in [MKQW25], the Lebesgue integral of the non-negative function vt_s is well-defined for all states $s \in S^{\mathcal{D}}$. With this function, we can now define the EVT of a state s as the expected value of vt_s over all possible paths.

Definition 3.18 (Expected Visiting Times in DTMCs)
For a state $s \in S$, the *expected visiting time* of s is characterized as

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

Additionally, we denote the *expected visiting time* that is obtained by fixing a certain state $t \in S$ as the only initial state as $\mathbb{E}^{\mathcal{D}}_t[vt_s]$. It is defined as

$$\mathbb{E}^{\mathcal{D}}_t[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}_t(\pi) \ .$$

Because the probability measure of paths π, see definition Def. 3.3, is defined as a product containing the factor $\iota_{init}(\pi[0])$, it is nonzero only for paths that begin in a state s with $\iota_{init}(s) > 0$. Thus, the value $\mathbb{E}^{\mathcal{D}}[vt_s]$ gives the number of times the state s is expected to be visited from all paths in $\mathcal{D}$ that start in one of the initial states. Whenever the DTMC $\mathcal{D}$ is evident from the context, we will omit the superscript $\mathcal{D}$ from $\mathbb{E}^{\mathcal{D}}[vt_s]$ and $\mathbb{E}^{\mathcal{D}}_t[vt_s]$.

Using the aforementioned partition of the set of states S into transient and recurrent states, see Def. 3.12, we can already give a characterization for the latter.

Lemma 3.19 (EVTs of Recurrent States)
For all recurrent states $s \in S_{re}$, it holds that

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if s is reachable,} \\ 0 & \text{else.} \end{cases}$$

For a proof of this lemma, see [MKQW25].

Thus, we will focus on finding the non-trivial EVTs of the transient states. For this, we will first define the *fundamental matrix* which was introduced by Kemeny and Snell [KS76].

Definition 3.20 (Fundamental Matrix)
For a DTMC $\mathcal{D}$ with the matrix $Q = (\mathbf{P}^{\mathcal{D}}(s,t))_{s,t \in S_{tr}}$ containing the transition probabilities between the transient states (defined in Def. 3.16), the *fundamental matrix* F is defined as,

$$F = (I - Q)^{-1} \ .$$

Kemeny and Snell show in [KS76] that Q converges to the null matrix. Therefore, the Neumann series of Q converges with $\sum_{k=0}^{\infty} Q^k = (\boldsymbol{I} - Q)^{-1}$, see Def. 2.11, i.e., the fundamental matrix is well-defined.

Importantly for this thesis, Kemeny and Snell show in [KS76], that the fundamental matrix gives for any two transient states $s, t \in S_{tr}$ the expected number of times s is visited, when starting in t. To be more precise, $\mathbb{E}_t[vt_s] = (F)_{ts}$. From this equation, it intuitively follows that transient states are only visited a finite amount of times.

Thus, we can use the fundamental matrix to characterize the EVTs for transient states. However, as the calculation of the inverse of a high-dimensional matrix becomes infeasible for large models, we will take a look at how the EVTs can be calculated using linear equation systems, as shown in [MKQW25].

First, we define

$$\tau_{init} = (\iota_{init}(s_1), \ldots, \iota_{init}(s_n))^{\mathsf{T}}$$

for all $s \in S_{tr}$ with which we can then introduce the following theorem taken from [MKQW25].

Theorem 3.21 (Linear Equation System Characterization of EVTs of Transient States)
The vector $(\mathbb{E}[vt_s])_{s \in S_{tr}}$ provides the unique solution of the following linear equation system in $\mathbb{R}^{|S_{tr}|}$:

$$(\boldsymbol{I} - Q^T) \cdot x = \tau_{init} \ .$$

For a proof of this theorem, see [MKQW25].

This equation system can then be rewritten as

$$x = \tau_{init} + Q^{\mathsf{T}} \cdot x \ .$$

For a single state $s \in S_{tr}$, we can thus calculate the respective EVT as follows,

$$\mathbb{E}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s) \cdot \mathbb{E}[vt_t] \ . \tag{3.2}$$

3.3 Relating Expected Visiting Times and Quantitative Reachability

We will now relate *expected visiting times* to *quantitave reachability* for DTMCs.

Lemma 3.22
Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define

$$(F)_{ts} = Pr_t(\Diamond s) \cdot (F)_{ss} \ ,$$

for all $s \neq t \in S_{tr}$, and thus

$$\begin{aligned} \mathbb{E}[vt_s] &= Pr(\Diamond s) \cdot (F)_{ss} && (3.3) \\ &= Pr(\Diamond s) \cdot \mathbb{E}_s[vt_s] \ . && (3.4) \end{aligned}$$

Trivially, all paths that reach s from t at least once visit s. Thus, intuitively, the *expected visiting time* of s starting in t is simply the product of the probability of reaching s and the *expected visiting time* of s starting in s.

Proof 2

$$\begin{aligned}
\mathbb{E}^{\mathcal{D}}[vt_s] &= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \exists k : \pi[k]=s} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi) \\
&+ \underbrace{\int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \nexists k : \pi[k]=s} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi)}_{=0} \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \exists k : \pi[k]=s} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi) \\
&= Pr(\Diamond s) \cdot \int_{\pi \in Paths^{\mathcal{D}}_{inf}} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}_s(\pi) \qquad (3.5)
\end{aligned}$$

□

???

Lemma 3.23
Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define

$$(F)_{ss} = \frac{1}{1 - Pr_s(\bigcirc \Diamond s)}$$

for all $s \in S_{tr}$, where $Pr_s(\bigcirc \Diamond s) = \sum_{t \in N(s)} \mathbf{P}(s,t) \cdot Pr_t(\Diamond s)$

(The fraction is well-defined as $Pr_s(\bigcirc \Diamond s) < 1$, see Def. 3.12)

Proof 3
As the only initial state is s, it is necessarily visited at least once. After that the probability that s will be visited at least one more time is $Pr_s(\bigcirc \Diamond s)$ and the probability that it will not, is accordingly $(1 - Pr_s(\bigcirc \Diamond s))$. Moreover, the probability of visiting s exactly k times is precisely the probability of returning $k-1$ times and not return again afterwards. Thus, $Pr_s(N_s = k) = Pr_s(\bigcirc \Diamond s)^{k-1}(1 - Pr_s(\bigcirc \Diamond s))$ for $k \in \mathbb{N}$, where N_s is a random variable representing the total number

of time s is visited (Pr_s means starting in s). Intuitively,

$$\begin{aligned}(F)_{ss} &= \sum_{k=1}^{\infty} k \cdot Pr_s(N_s = k) \\ &= \sum_{k=1}^{\infty} k \cdot Pr_s(\bigcirc\Diamond s)^{k-1}(1 - Pr_s(\bigcirc\Diamond s)) \\ &= (1 - Pr_s(\bigcirc\Diamond s)) \sum_{k=1}^{\infty} k \cdot Pr_s(\bigcirc\Diamond s)^{k-1} \\ &= (1 - Pr_s(\bigcirc\Diamond s)) \cdot \frac{1}{(1 - Pr_s(\bigcirc\Diamond s))^2} = \frac{1}{1 - Pr_s(\bigcirc\Diamond s)} \qquad \text{(by geometric series.)}\end{aligned}$$

□

Finally, using **??**,

$$\begin{aligned}\mathbb{E}[vt_s] &= Pr(\Diamond s) \cdot (F)_{ss} && \text{(by Lemma 3.22)} \\ &= \frac{Pr(\Diamond s)}{1 - Pr_s(\bigcirc\Diamond s)} && \text{(by Lemma 3.23)} \qquad (3.6) \\ &= \sum_{t \in S_{tr} \cap in} \frac{Pr_t(\Diamond s)}{1 - Pr_s(\bigcirc\Diamond s)} \cdot \iota_{init}(t)\end{aligned}$$

Notice, that for any given DTMC $\mathcal{D}$ we can construct $\mathcal{D}^s_{uni} = (S^{\mathcal{D}} \cup \{s_{in}, s_{suc}\}, \mathbf{P}^{\mathcal{D}^s_{uni}}, \iota^{\mathcal{D}^s_{uni}}_{init})$, where $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_i, s_j) = \mathbf{P}^{\mathcal{D}}(s_i, s_j)$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s, s_j) = 0$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s, s_{suc}) = 1$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{suc}, s_j) = \mathbf{P}^{\mathcal{D}}(s, s_j)$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{in}, s_j) = \iota^{\mathcal{D}}_{init}(s_j)$ for all $s_i \in S^{\mathcal{D}} \backslash \{s\}$ and $s_j \in S^{\mathcal{D}}$, and $\iota^{\mathcal{D}^s_{uni}}_{init}(s_{in}) = 1$. Essentially, $\mathcal{D}^s_{uni}$ is a copy of $\mathcal{D}$, where instead of using ι_{init} as an initial distribution, we simply have a single initial state s_{in} and s has only one direct successor s_{suc}, which has the same probability distribution as s in $\mathcal{D}$.

Theorem 3.24
The reachability probabilities and thus also EVTs in $\mathcal{D}$ and the corresponding $\mathcal{D}^s_{uni}$ are identical.

Proof 4
First, we show that the reachability probabilities are equal (see Def. 3.5) $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}^s_{uni}}(\Diamond T)$ for all $T \in S^{\mathcal{D}}$. For this, we define the bijection $\phi : Paths^{\mathcal{D}^s_{uni}}_{inf} \to Paths^{\mathcal{D}}_{inf}, \pi = (s_{in}, s_1, s_2, \dots, s_{k-1}, s_{suc}, s_k, \dots) \mapsto (s_1, s_2, \dots, s_{k-1}, s_k, \dots)$, i.e., the bijection ϕ removes s_{in} and all s_{suc}. This bijection is well defined, as $N(s_{in}) = \{s \mid \iota^{\mathcal{D}}_{init}(s) > 0\}$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_i, s_{suc}) > 0$ only for $s_i = s$, and $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{suc}, s_i) = \mathbf{P}^{\mathcal{D}}(s, s_i)$ ($\to$ thus same (amount of) paths).

Moreover, assume a given $\pi \in Paths_{inf}^{\mathcal{D}_{uni}}$ has $\pi[k_i] = s_{suc}$ with $i \in \underline{j}$ for some $j \in \mathbb{N}_{>0}$ and $\phi(\pi) = \hat{\pi}$.

$$
\begin{aligned}
Pr^{\mathcal{D}_{uni}}(\pi) &= \underbrace{\iota_{init}^{\mathcal{D}_{uni}}(\pi[0])}_{=1} \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \\
&= \mathbf{P}^{\mathcal{D}_{uni}}(\pi[0], \pi[1]) \cdot \underbrace{\prod_{i=1}^{j-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[\underbrace{k_i - 1}_{=s}], \pi[\underbrace{k_i}_{=s_{suc}}])}_{=1} \cdot \prod_{k=1}^{k_1-2} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \\
&\quad \cdot \prod_{k=k_1}^{k_2-2} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \cdot \ldots \cdot \prod_{k_j}^{|\pi|-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k = k_j], \pi[k+1]) \\
&= \iota_{init}^{\mathcal{D}}(\phi(\pi)[1]) \cdot \prod_{k=1}^{|\hat{\pi}|-1} \mathbf{P}^{\mathcal{D}}(\phi(\pi)[k], \phi(\pi)[k+1]) \qquad (\text{as, } \pi[k_i+1] = \hat{\pi}[k_i]) \\
&= \iota_{init}^{\mathcal{D}}(\hat{\pi}[0]) \cdot \prod_{k=0}^{|\hat{\pi}|-1} \mathbf{P}^{\mathcal{D}}(\hat{\pi}[k], \hat{\pi}[k+1]) \\
&= Pr^{\mathcal{D}}(\hat{\pi}) \ .
\end{aligned}
$$

Therefore, we can conclude $Pr^{\mathcal{D}_{uni}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond T)$, as $\Diamond T \subseteq Paths_{inf}^{\mathcal{D}_{uni}}$. Using Eq. (3.6), we can then also conclude that the EVTs of $\mathcal{D}$ and $\mathcal{D}_{uni}$ must be equal.□

Thus, the reachability probabilities $Pr_{s_{in}}^{\mathcal{D}_{uni}}(\Diamond s) = \sum_{t \in S_{in}} Pr_t^{\mathcal{D}}(\Diamond s) = \sum_{t \in S_{tr \cap in}} Pr_t^{\mathcal{D}}(\Diamond s)$ are equal and we can simplify the previous expression,

$$
\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr_{s_{in}}^{\mathcal{D}_{uni}}(\Diamond s)}{1 - Pr_{s_{suc}}^{\mathcal{D}_{uni}}(\Diamond s)} \cdot \tau_{init}(s_{in}) \left(= \frac{Pr^{\mathcal{D}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}}(\bigcirc \Diamond s)} \cdot \tau_{init}(s_{in}) \right) \ . \tag{3.7}
$$

In other words, we can calculate the EVT of a DTMC $\mathcal{D}$ with only two reachability probabilities using the corresponding DTMC $\mathcal{D}_{uni}$.

Moreover, using Lemma 3.22 and the construction $\mathcal{D}_{uni}$, we can calculate the quantitative reachability of a given state with $Pr(\Diamond s) = \frac{(F^{\mathcal{D}_{uni}})_{s_{in}s}}{(F)_{ss}}$.

Chapter 4

Interval Markov Chains

The focus of this thesis are expected visiting times in Interval Markov Chains, which we will introduce in this chapter. In the following sections we will mainly discuss how the expected visiting times can be calculated and shortly examine stationary distributions of Interval Markov Chains.

We start by introducing concepts which we will use in the following sections.

Definition 4.1 (Interval Markov Chain)
An interval discrete-time Markov chain (IDTMC) is a tuple $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$, where

(i) $S^{\mathcal{I}}$ is a countable, nonempty set of states,

(ii) $\underline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the lower bound on the transition probability function, such that $\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \leq \mathbf{P}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$,

(iii) $\overline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the upper bound on the transition probability function, such that $\mathbf{P}(s_i, s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$,

(iv) $\underline{\iota}^{\mathcal{I}}_{init} : S^{\mathcal{I}} \to [0,1]$ is the lower bound on the initial distribution, such that $\underline{\iota}^{\mathcal{I}}_{init}(s) \leq \iota_{init}(s)$ for all $s \in S^{\mathcal{I}}$, and

(v) $\overline{\iota}^{\mathcal{I}}_{init} : S \to [0,1]$ is the upper bound on the initial distribution, such that $\iota_{init}(s) \leq \overline{\iota}^{\mathcal{I}}_{init}(s)$ for all $s \in S^{\mathcal{I}}$.

Whenever the IDTMC $\mathcal{I}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{I}}$, and $\iota^{\mathcal{I}}_{init}$. In contrast to the precise transition probability function and initial distribution of DTMCs, Interval Markov Chains allow for uncertainty.

Example 4.2
The Fig. 4.1 depicts an exemplary IDTMC $\mathcal{I}_E = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, where $S = \{s_1, s_2, s_3, s_4, s_5\}$, the lower and upper transition probability functions are given by

$$\underline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.2 \\ 0.2 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \overline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0.8 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the lower and upper initial distributions

$$\underline{\iota}_{init} = (0.5, 0, 0, 0, 0)^\mathsf{T} \qquad \overline{\iota}_{init} = (1, 0, 0.5, 0, 0)^\mathsf{T}.$$

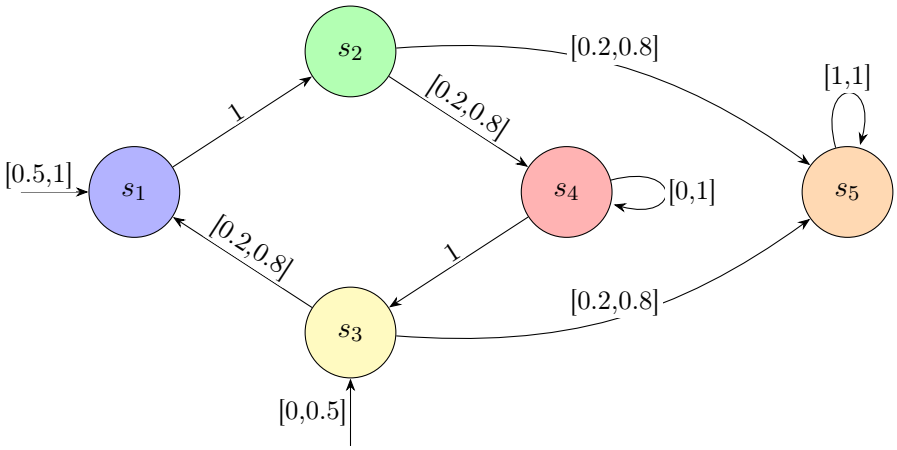


Figure 4.1: IDTMC $\mathcal{I}$

Definition 4.3 (Path of an IDTMC)
Similarly to DTMCs, an *infinite path* in an IDTMC $\mathcal{I} = (S,)$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S^{\mathcal{I}}$, such that $\underline{\mathbf{P}}(\pi[k], \pi[k+1]) > 0$ for all $k \in \mathbb{N}$.

Fixed-initial-state definition. In the literature, see, e.g., [CHK13], it is common to represent the initial distribution by an additional initial state rather than introducing a separate initial-distribution function. In this case, an IMC is given as a tuple $\mathcal{I}_{\text{uni}} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, s_{\text{in}})$, where s_{in} is the unique initial state. We refer to this as the *fixed-initial-state definition*.

More precisely, let $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$ be an IMC in our definition, and let $\iota^{\mathcal{I}}_{\text{init}}$ denote its initial distribution. The corresponding fixed-initial-state representation is obtained by introducing a fresh state s_{in} and defining $S^{\mathcal{I}_{\text{uni}}} = S^{\mathcal{I}} \cup \{s_{\text{in}}\}$.

The lower and upper transition-probability functions are given by

$$\underline{\mathbf{P}}^{\mathcal{I}_{\text{uni}}}(s_i, s_j) = \begin{cases} \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), & s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{I}}_{\text{init}}(s_j), & s_i = s_{\text{in}}, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\overline{\mathbf{P}}^{\mathcal{I}_{\text{uni}}}(s_i, s_j) = \begin{cases} \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), & s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{I}}_{\text{init}}(s_j), & s_i = s_{\text{in}}, \\ 0, & \text{otherwise}. \end{cases}$$

The same construction applies to DTMCs. Let $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota^{\mathcal{D}}_{\text{init}})$ be a DTMC. Its corresponding fixed-initial-state representation is $\mathcal{D}_{\text{uni}} = (S^{\mathcal{D}} \cup \{s_{\text{in}}\}, \mathbf{P}^{\mathcal{D}_{\text{uni}}}, s_{\text{in}})$, where

$$\mathbf{P}^{\mathcal{D}_{\text{uni}}}(s_i, s_j) = \begin{cases} \mathbf{P}^{\mathcal{D}}(s_i, s_j), & s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{D}}_{\text{init}}(s_j), & s_i = s_{\text{in}}, \\ 0, & \text{otherwise}. \end{cases}$$

The two representations induce the same probability space up to the natural bijection between their paths. In particular, define

$$\phi : Paths^{\mathcal{D}_{\text{uni}}}_{\text{inf}} \longrightarrow Paths^{\mathcal{D}}_{\text{inf}}, (s_{\text{in}}, s_0, s_1, \ldots) \mapsto (s_0, s_1, \ldots).$$

ϕ is a bijection, since s_{in} has no incoming transitions, every infinite path of $\mathcal{D}_{\text{uni}}$ starts with s_{in}.

To see that the probability measures correspond under ϕ, consider a finite path $\pi_{\text{uni}} = (s_{\text{in}}, s_0, \ldots, s_n)$ of $\mathcal{D}_{\text{uni}}$, and let $\pi = (s_0, \ldots, s_n)$ be the corresponding finite path of $\mathcal{D}$. Then

$$Pr^{\mathcal{D}_{\text{uni}}}(\pi_{\text{uni}}) = \mathbf{P}^{\mathcal{D}_{\text{uni}}}(s_{\text{in}}, s_0) \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}_{\text{uni}}}(s_k, s_{k+1}) \quad = \iota^{\mathcal{D}}_{\text{init}}(s_0) \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}}(s_k, s_{k+1}) = Pr^{\mathcal{D}}(\pi).$$

Thus, corresponding cylinder sets have the same probability, and consequently the probability measures on the two path spaces coincide under the bijection ϕ.

Hence, adding a unique initial state is merely a different representation of the initial distribution and does not change the probabilistic behaviour of the model. In particular, results formulated for the fixed-initial-state definition can be transferred to our definition by applying the above construction.

To resolve the uncertainty of the initial distribution and transition probability function, we need to define semantics to interpret the Interval Markov Chain. In other words, we need to resolve an Interval Markov Chain to some set of Markov Chains that correspond to it. We say that all of these DTMCs $\mathcal{D}$ satisfy the IDTMC $\mathcal{I}$ and denote it as $\mathcal{D} \models \mathcal{I}$. We assume in this thesis that there exists at least one satisfying DTMC for every IDTMC.

Assumption 1
In this thesis, we only consider finite IDTMCs, i.e., IDTMCs that have a finite state space. Moreover, we will only consider IDTMCs where there exists at least one satisfying DTMC $\mathcal{D} \models \mathcal{I}$ for each state $s \in S^{\mathcal{I}}$, such that s is reachable with a positive probability from an initial state in $\mathcal{D}$. Because, if a given state was not reachable in any satisfying DTMC, then the EVT of this state would trivially be 0 for all satisfying DTMCs and have no impact on the EVTs of other states.

In this thesis, we will consider two popular semantics: the *once-and-for-all*, also known as the *Uncertain Markov Chain (UMC) semantic*, and the *IMDP semantic*. We will first define the former, similar to [BDLM21, CSH08, BLW13]. To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \bar{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$.

Definition 4.4 (Once-and-for-all/UMC Semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ if and only if $S^{\mathcal{I}} = S^{\mathcal{D}}$, and for all $s, s' \in S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{D}}(s, s') \in [\underline{\mathbf{P}}^{\mathcal{I}}(s, s'), \overline{\mathbf{P}}^{\mathcal{I}}(s, s')]$ and $\iota^{\mathcal{D}}_{init}(s) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \bar{\iota}^{\mathcal{I}}_{init}(s)]$.

In other words, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC is satisfactory. Thus, we can interpret the *UMC* semantic for an IDTMC, as the initial distribution and probability function being chosen from the intervals *once-and-for-all*. Moreover, notice that $\mathcal{D}$ has the same cardinality as $\mathcal{I}$, i.e., $|S^{\mathcal{D}}| = |S^{\mathcal{I}}|$. If $\mathcal{D}$ satisfies $\mathcal{I}$ under the UMC semantic, we denote this as $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$.

Example 4.5
Consider the IDTMC $\mathcal{I}_E$ in Fig. 4.1. The DTMC $\mathcal{D}_U = (S^{\mathcal{I}_E}, \mathbf{P}, \iota_{init})$ depicted in Fig. 4.2 satisfies $\mathcal{I}$ under the UMC semantic, i.e., $\mathcal{D}_U \models_U \mathcal{I}$, where the transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the initial distribution

$$\iota_{init} = (1, 0, 0, 0, 0)$$

The second semantic which we will define, is the *IMDP semantic*. There are two common definitions for this semantic. In this thesis, we will use the definition found in [SVA06] and [CSH08].

While the *UMC* semantic selects a particular probability function *once-and-for-all*, the *IMDP* semantic allows for a probability function to be picked from the set of valid

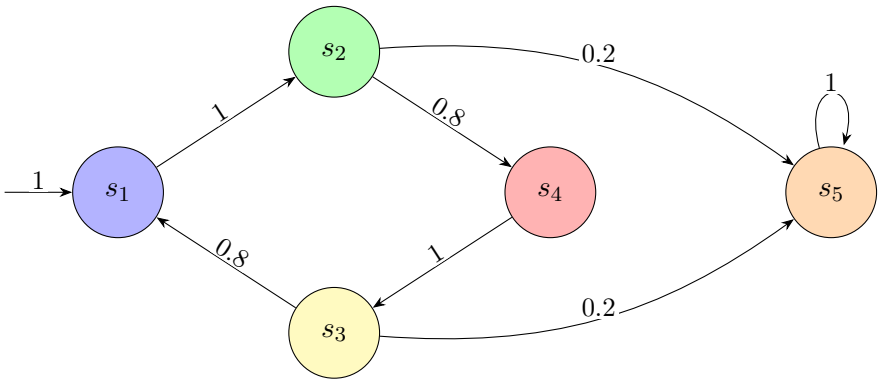


Figure 4.2: DTMC $\mathcal{D}_U$ with $\mathcal{D}_U \models_U \mathcal{I}$

functions before every transition. Similar to [SVA06] and [CSH08], we define the set of valid probability functions over $S^{\mathcal{I}}$ as $Steps^{\mathcal{I}}(s)$, i.e. $Steps^{\mathcal{I}}(s) = \{\mathbf{P} \mid \sum_{t \in S^{\mathcal{I}}} \mathbf{P}(s,t) = 1 \text{ and } \underline{\mathbf{P}}(s,t) \leq \mathbf{P}(s,t) \leq \overline{\mathbf{P}}(s,t) \text{ for all } t \in S^{\mathcal{I}}\}$ for all $s \in S^{\mathcal{I}}$. Furthermore, $Steps^{\mathcal{I}} = \bigcup_{s \in S^{\mathcal{I}}} = Steps^{\mathcal{I}}(s)$. In other words, under the *IMDP* semantic, a potentially different probability function can be chosen from $Steps^{\mathcal{I}}(s)$ before each time the state s is entered. This choice of a probability function is made non-deterministically by an external environment.

In [Bad] this external environment is defined as the *nature*.

Definition 4.6 (Nature)
A *nature* for an IDTMC $\mathcal{I}$ is a function $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ that assigns each finite path to a valid transition probability function, i.e., $\eta(\pi) \in Steps^{\mathcal{I}}(last(\pi))$.

Notice, that we only define this external environment for the transition probability function and not the initial distribution. Recall, that results for the *fixed-initial-state definition* can be applied to our definition by applying the shown construction. Thus, it suffices to define this external environment for the transition probability function.

By applying a given *nature* η to $\mathcal{I}$, we resolve the uncertainty of the probability function, as η specifies a probability function $\mathbf{P} \in Steps^{\mathcal{I}}(s)$ for every finite path in $\mathcal{I}$. Thus, η induces a DTMC as follows. The definition is based on [BK08].

Definition 4.7 (DTMC of an IDTMC Induced by a Nature)
For an IDTMC $\mathcal{I}$ and a nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (Paths^{\mathcal{I}}_{fin}, \mathbf{P}^{\mathcal{D}_\eta}, \iota^{\mathcal{I}}_{init}) \ ,$$

where for all $\pi \in Paths^{\mathcal{I}}_{fin}$, $s \in S^{\mathcal{I}}$, and $\eta(\pi) = \mathbf{P}$:

$$\mathbf{P}^{\mathcal{D}_\eta}(\pi, \pi') = \begin{cases} \mathbf{P}(last(\pi), s) & \pi' = (\pi[0], \pi[1], \ldots, last(\pi), s) \\ 0 & \text{otherwise.} \end{cases}$$

Notice, that these induced DTMCs are infinite in size, i.e., $|\mathcal{D}_\eta| = \infty$. To relate the states in the induced DTMCs $\mathcal{D}_\eta$ to the states in the IDTMC $\mathcal{I}$, we define the function

$$\tau : Paths^{\mathcal{I}}_{fin} \to S^{\mathcal{I}},\ (\pi[0], \ldots, last(\pi)) \mapsto last(\pi)\ .$$

In other words, the states in the induced DTMCs that correspond to paths ending in some state $s \in S^{\mathcal{I}}$, are mapped to s. Intuitively,

$$\tau^{-1}(s) = \{\pi \in Paths_{fin} \mid \tau(\pi) = s\}$$

defines the set of all states in the induced DTMCs that correspond to paths ending in s for all $s \in S^{\mathcal{I}}$.

Using these notions, we can now define the *IMDP semantic*.

Definition 4.8 (IMDP semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ if and only if it is induced by a *nature* $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ on $\mathcal{I}$.

If $\mathcal{D}$ satisfies $\mathcal{I}$ under the IMDP semantic, we denote this as $\mathcal{D} \models_{\mathrm{I}} \mathcal{I}$.

An important type of nature, which we require in this thesis, is the *stationary nature*. The following definition is based on [Bad].

Definition 4.9 (Stationary Nature)
A nature η for an IDTMC $\mathcal{I}$ is *stationary* if for all paths $\pi, \pi' \in Paths^{\mathcal{I}}_{fin}$, it holds that

$$last(\pi) = last(\pi') \implies \eta(\pi) = \eta(\pi')\ .$$

Intuitively, this implies that η only depends on the current state of the model, i.e., the state the model is in. Thus, stationary natures are functions of the form

$$\eta : S^{\mathcal{I}} \to Steps^{\mathcal{I}}\ .$$

Notice that the induced DTMCs of these stationary natures are finite in size.

Definition 4.10 (DTMC of an IDTMC Induced by a Stationary Nature)
For an IDTMC $\mathcal{I}$ and a stationary nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_\eta}, \iota^{\mathcal{I}}_{init})\ ,$$

where for all $s_i \in S^{\mathcal{I}}$ and $\eta(s_i) = \mathbf{P}$:

$$\mathbf{P}^{\mathcal{D}_\eta}(s_i, s_j) = \begin{cases} \mathbf{P}(s_i, s_j) & s_j \in N(s_i), \\ 0 & \text{otherwise.} \end{cases}$$

The other common definition can be found in [BDL+17] and [BDLM21].

Example 4.11
Consider the IDTMC $\mathcal{I}$ in Fig. 4.1. The DTMC $\mathcal{D}_I$ depicted in Fig. 4.3 satisfies $\mathcal{I}$ under the IMDP semantic, i.e., $\mathcal{D}_I \models_I \mathcal{I}$, where the transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the initial distribution

$$\iota_{init} = (0.6, 0, 0, 0, 0)$$

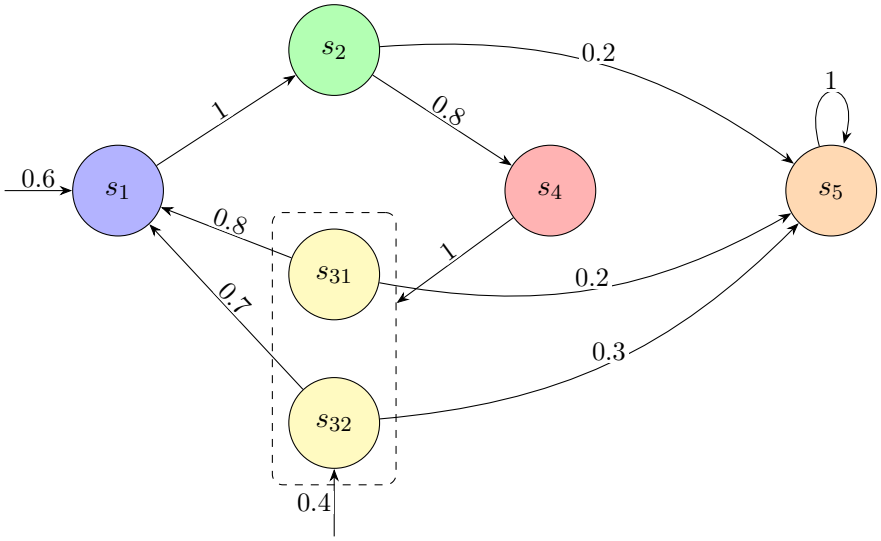


Figure 4.3: DTMC $\mathcal{D}_I$ with $\mathcal{D}_I \models_I \mathcal{I}$.

If we want to emphasize that a DTMC $\mathcal{D}$ satisfies the IDTMC $\mathcal{I}$ under both semantics, we write $\mathcal{D} \models \mathcal{I}$. Furthermore, if two DTMCs $\mathcal{D}_1$ and $\mathcal{D}_2$ satisfy a given IDTMC $\mathcal{I}$, we write $\mathcal{D}_1, \mathcal{D}_2 \models \mathcal{I}$ for short.

Lastly, we introduce some of notions of IDTMCs, which we use in the following chapter.

First, we introduce reachability probabilities for IDTMCs. We use the definition of the quantitative reachability property in IDTMCs provided in [BDL+17].

Definition 4.12 (Reachability Probability in IDTMCs)
Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, then a quantitative reachability property on $\mathcal{I}$ is defined as $Pr^{\mathcal{I}}(\Diamond T) \sim p$, where $0 \leq p \leq 1$ and $\sim \in \{\leq, <, >, \geq\}$. This quantitative property is satisfied if and only if there exists at least one DTMC $\mathcal{D} \models \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond T) \sim p$.

In this thesis, we are particularly interested in the optimal reachability probabilities, which we define as follows.

Definition 4.13 (Maxmimum and Minimum Reachability Probability)
The maximum reachability $\ldots \overline{Pr}^{\mathcal{I}}(\Diamond T) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$

To shorten the notation we will denote $\overline{Pr}_t^{\mathcal{D}}(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$ and $\underline{Pr}_t^{\mathcal{D}}(\Diamond s) = \inf_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$.

Furthermore, transient and recurrent states are defined as follows.

Definition 4.14 (Transient and Recurrent states in IDTMCs)
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}$ is called recurrent if and only if $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1$.

Furthermore, a state $s \in S^{\mathcal{I}}$ is called transient if and only if $\underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) < 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) < 1$.

We denote the set of recurrent states as $S_{re}^{\mathcal{I}}$ and the set of transient states as $S_{tr}^{\mathcal{I}}$.

Notice, that in contrast to DTMCs, there exist IDTMCs $S_{tr}^{\mathcal{I}} \cap S_{re}^{\mathcal{I}} \neq \emptyset$. To observe this, consider state s_4 of Example 4.2. Let $\mathcal{D}_{tr} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{tr}}(s_4, s_3) = 1$. Therefore, $s_4 \in S_{tr}^{\mathcal{D}_{tr}}$ and consequently $s_4 \in S_{tr}^{\mathcal{I}_E}$. Now, let $\mathcal{D}_{re} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{re}}(s_4, s_4) = 1$. Then, $s_4 \in S_{re}^{D_{re}}$ and thus $s_4 \in S_{re}^{\mathcal{I}_E}$. Thus, $s_4 \in S_{re}^{\mathcal{I}_E} \cap S_{tr}^{\mathcal{I}_E}$.

We will (indirectly??) show in the following Sec. 4.0.1, that $\mathcal{D}_{tr} \models_I \mathcal{I}_E$. Thus, ...

Lastly, BSCCs are defined as follows.

Definition 4.15 (BSCCs for IDTMCs)
Let $\mathcal{I}$ be an IDTMC, then $B^{\mathcal{D}} \subseteq S^{\mathcal{I}}$ is a BSSC, if $B^{\mathcal{D}}$ is a BSCC in the DTMC $\mathcal{D}$, see Def. 3.10, and $\mathcal{D} \models \mathcal{I}$.

4.0.1 Relation between the semantics

Intuitively, the *IMDP* semantic is more expressive than the *UMC* semantic. More specifically for any given IDTMC $\mathcal{I}$, a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$ according to the UMC semantic, i.e., $\mathcal{D} \models_U \mathcal{I}$, does also satisfy $\mathcal{I}$ according to the IMDP semantic, i.e., $\mathcal{D} \models_I \mathcal{I}$.

Theorem 4.16
For any IDTMC $\mathcal{I}$, if $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$.

We can prove this theorem using the following lemma.

Lemma 4.17
Given an IDTMC $\mathcal{I}$ and stationary nature η, for every induced DTMC $\mathcal{D}_\eta$, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$, such that $\mathcal{D} = \mathcal{D}_\eta$.

In other words the set of induced DTMCs by stationary natures on $\mathcal{I}$ is identical to the set of DTMCs that satisfy $\mathcal{I}$ under the UMC semantic.

Proof 5
First notice, that $S^{\mathcal{D}_\eta} = S^{\mathcal{I}}$ by definition, see Def. 4.10. □

As stationary natures are a specific type of nature, Theorem 4.16 follows directly from Lemma 4.17.

Importantly, the two semantics have the same maximum and minimum reachability probability for any given (initial) state, according to [BDL+17] and [CHK13]. ...

4.1 Expected Visiting Times in IDTMCs

We continue by defining *expected visiting times* (EVTs) for IDTMCs. As we discussed in the previous section, IDTMCs can be seen as a set of DTMCs that satisfy the IDTMC according to the given semantic. These different DTMCs can have varying EVTs for a given state s in the IDTMC, thus we are interested in the upper and lower bound for the EVTs of s, i.e., in the highest and lowest EVT of s among the satisfying DTMCs.

Definition 4.18 (EVTs of IDTMCs)
For a IDTMC $\mathcal{I}$, we define the *upper expected visiting time* (uEVT) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and *lower expected visiting time* (ℓEVT) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ for all states $s \in S^{\mathcal{I}}$, as follows for the UMC semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

and the IMDP semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_I \mathcal{I}} \int_{\tau^{-1}(s)} \mathbb{E}^{\mathcal{D}}[vt_t]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_I \mathcal{I}} \int_{\tau^{-1}(s)} \mathbb{E}^{\mathcal{D}}[vt_t]$

4.1.1 Relating Expected Visiting Times and Quantitative Reachability

Similar to Sec. 3.3, we now relate *expected visiting times* to *quantitative reachability*. Moreover, we show that for any IDTMC $\mathcal{I}$, there exists a DTMC $\mathcal{D} \models \mathcal{I}$ that has the optimal, i.e., maximum or minimum, reachability probability for all transient states in $\mathcal{I}$.

Given the quantitative reachabilities of the DTMCs that satisfy a certain IDTMC $\mathcal{I}$, we can calculate the corresponding EVTs of a given state s in $\mathcal{I}$ and thus find the lEVTs and uEVTs using the results from Sec. 3.3.

As we concluded in Sec. 3.3, we can calculate the EVT for all transient states $s \in S^{\mathcal{D}}_{tr}$ in a DTMC $\mathcal{D}$ via quantitative reachability, using the corresponding DTMC $\mathcal{D}_{uni}$ and the Eq. (3.6) $\mathbb{E}^{\mathcal{D}}[vt_s] = \sum_{t \in S_{tr} \cap in} \frac{Pr^{\mathcal{D}}_t(\Diamond s)}{1 - Pr^{\mathcal{D}}_s(\bigcirc \Diamond s)} \cdot \iota_{init}(t)$. As the uEVT and ℓEVT, under the UMC semantic, are given by the maximum and minimum EVT

Thus, the uEVT and ℓEVT of a given IDTMC $\mathcal{I}$ are given by the largest and smallest ratio of these quantitative reachability probabilites among the satisfying Markov Chains.

Remark. *Existence of DTMC with optimal reachability.* For every transient state $t \in S_{tr}$ there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ with $Pr^{\mathcal{D}}_t(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr^{\mathcal{D}_i}_t(\Diamond s)$. This follows directly from the compactness and continuity of the polytope of DTMCs satisfying $\mathcal{I}$; it is not reproved here, only reused. To shorten the notation we will denote $\overline{Pr}^{\mathcal{D}}_t(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr^{\mathcal{D}_i}_t(\Diamond s)$ and $\underline{Pr}^{\mathcal{D}}_t(\Diamond s) = \inf_{\mathcal{D}_i \models_U \mathcal{I}} Pr^{\mathcal{D}_i}_t(\Diamond s)$.

Theorem 4.19
Given an IDTMC $\mathcal{I}$, a transient state $t \in S^{\mathcal{I}}$, a state $s \in S^{\mathcal{I}}$, and a DTMCs $\mathcal{D}$ with $\mathcal{D} \models \mathcal{I}$. (Positive transition/reachability probability from t to s, i.e. there needs to exist a path from t to s)

If $\mathcal{D}$ has the maximum reachability probability from t to s, i.e., $Pr^{\mathcal{D}}_t(\Diamond s) = \overline{Pr}^{\mathcal{I}}_t(\Diamond s)$, then for all $u \in I^{\mathcal{D}}_{t,s}$:

$$Pr^{\mathcal{D}}_u(\Diamond s) = \overline{Pr}^{\mathcal{I}}_u(\Diamond s)$$

Proof 6
We will start by proving this theorem for $\mathcal{D} \models_I \mathcal{I}$ and then show why $\mathcal{D} \models_U \mathcal{I}$ follows from it.

As shown in [BK08, Theorem 10.100], for any MDP $\mathcal{M}$ the maximum reachability probability is characterized by the equation

$$\overline{Pr}_t^{\mathcal{I}}(\Diamond s) = \max \left\{ \sum_{u \in N(t)} \mathbf{P}(t, \alpha, u) \cdot \overline{Pr}_u^{\mathcal{I}}(\Diamond s) \mid \alpha \in Act(t) \right\} . \tag{4.1}$$

□

Clearly, the policies which solve this equation have the maximum reachability probability $Pr_u(\Diamond s)$ for all $u \in I_{t,s}^{\mathcal{D}}$, as shown in [BK08]. As shown in the previous section, an IDTMC under the IMDP semantic can be mapped to a MDP, where the policy represents a given satisfying DTMC (bzw. nach einer/der üblichen IMDP Semantiken ist eine policy/adversary das satisfying DTMC). Thus, this theorem is proven for the IMDP semantic. (Also importantly, the solution to the set of equations is unique, as shown in [BK08]. Thus, the maximum reachability probabilities of the states are unique)

As proven in [BK08] (Lemma 10.102 and Lemma 10.113), given any finite MDP $\mathcal{M}$ with state space S, and $T \subseteq S$, there exists *memoryless* schedulers $\mathfrak{S}_{max}$ and $\mathfrak{S}_{min}$ for every $t \in S$, such that $\overline{Pr}_t^{\mathcal{M}}(\Diamond T) = Pr_t^{\mathfrak{S}_{max}}(\Diamond T)$ and $\underline{Pr}_t^{\mathcal{M}}(\Diamond T) = Pr_t^{\mathfrak{S}_{min}}(\Diamond T)$. In other words, for every MDP, there exist *memoryless* schedulers that has maximum or minimum reachability to T from t. As discussed, a memoryless scheduler on the MDP construction induces a DTMC that satisfies the IDTMC. Because this memoryless scheduler is optimal, clearly it satisfies Eq. (4.1). Thus, this theorem is proven for the UMC semantic.

Corollary 4.20
Given an IDTMC $\mathcal{I}$, two transient states $t_1, t_2 \in S^{\mathcal{I}}$, a (goal) state $s \in S^{\mathcal{I}}$, and two DTMCs $\mathcal{D}_1, \mathcal{D}_2$ with $\mathcal{D}_1, \mathcal{D}_2 \models_U \mathcal{I}$ and $Pr_{t_1}^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}(\Diamond s)$ and $Pr_{t_2}^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_{t_2}^{\mathcal{I}}(\Diamond s)$. In other words the DTMCs have a maximum reachability probability for t_1 and t_2, respectively.

For all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$, the reachability probabilities $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ are identical.

Proof 7
This corollary follows from Theorem 4.19. Using Theorem 4.19, we can conclude that $Pr_u^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1}$ and $Pr_u^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_2,s}^{\mathcal{D}_2}$. Thus, $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$. □

Lemma 4.21 (One-state Extension)
Given a set $T \subset S$, a state $t \notin T$, and a (goal) state s, assume that $\mathcal{D}_T \models_U \mathcal{I}$ with $Pr_u^{\mathcal{D}_T}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $\mathcal{D}_t \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. Define $\mathcal{D}_{cmb} \models_U \mathcal{I}$ by

$$\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \begin{cases} \mathbf{P}^{\mathcal{D}_t}(w,v) & \text{if } w \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T} \\ \mathbf{P}^{\mathcal{D}_T}(w,v) & \text{else.} \end{cases}$$

Stated in words, $\mathcal{D}_{cmb}$ is constructed such that its probability distribution function is equal to that of $\mathcal{D}_t$ in all states that are only intermediate states for t and not T, while it is equal to that of $\mathcal{D}_T$ in all other states.

Then, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T \cup \{t\}$, i.e., $\mathcal{D}_{cmb}$ has the maximum reachability probability for all states in $T \cup \{t\}$.

Proof 8
As $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_T}(w,v)$ for all $w \in I_{T,s}^{\mathcal{D}_T}$, clearly $\overline{Pr}_u^{\mathcal{I}}(\Diamond s) = Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T$, see Lemma 3.8.

Thus, we need to show that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. As we know that $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, we will prove that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_t^{\mathcal{D}_t}(\Diamond s)$. Notice, that if $I_{t,s}^{\mathcal{D}_t} \cap I_{T,s}^{\mathcal{D}_T} = \emptyset$, then the proof concludes using the same argument as for $u \in T$. Furthermore, if $I_{t,s}^{\mathcal{D}_t} \subseteq I_{T,s}^{\mathcal{D}_T}$, then the proof concludes directly using Theorem 4.19.

Recall, that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \sum_{u \in N(t) \cap I_{t,s}^{\mathcal{D}_{cmb}}} \mathbf{P}^{\mathcal{D}_{cmb}}(t,u) \cdot Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$, see Lemma 3.8. For all $u \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T}$, $\mathbf{P}^{\mathcal{D}_{cmb}}(u,w) = \mathbf{P}^{\mathcal{D}_t}(u,w)$ and for all $u \in I_{T,s}^{\mathcal{D}_T} \cap I_{t,s}^{\mathcal{D}_t}$, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_t}(\Diamond s)$, see Corollary 4.20. From this, we can conclude that $Pr_t^{\mathcal{D}_t}(\Diamond s) = Pr_t^{\mathcal{D}_{cmb}}(\Diamond s)$. Thus, we have shown that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Into only remains to show, that $\mathcal{D}_{cmb} \models_U \mathcal{I}$. As $\mathcal{D}_t, \mathcal{D}_T \models_U \mathcal{I}$ it clearly follows from the construction that also $\mathcal{D}_{cmb} \models \mathcal{I}$. □

Crucially, Lemma 4.21 can be extended to all transient states.

Theorem 4.22
Given an IDTMC $\mathcal{I}$ and a (goal) state $s \in S$ there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_t^{\mathcal{D}}(\Diamond s) = \sup_{\mathcal{D}_i \models_U \mathcal{I}} Pr_t^{\mathcal{D}_i}(\Diamond s)$ for all transient states $t \in S_{tr}^{\mathcal{I}}$.

Proof 9
Consider any given enumeration $t_1, t_2, \ldots$ of $S^{\mathcal{I}}$. Clearly, this enumeration is finite,

since $S^{\mathcal{I}}$ is finite, see **??**.

Construction. Now, let $\mathcal{D}^{(1)} \models_U \mathcal{I}$ with $Pr_{t_1}^{\mathcal{D}^{(1)}}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}$ and $T_1 := I_{t_1,s}^{\mathcal{D}^{(1)}}$, trivially such a DTMC must exist. By Theorem 4.19, $Pr_u(\Diamond s) = \overline{\mathbf{P}}_u(\Diamond s)$.

Inductively, $\mathcal{D}^{(n)} \models_U \mathcal{I}$ (and $s \in T_n$). If $t_{n+1} \in T_n$, then we set $\mathcal{D}^{(n+1)} := \mathcal{D}^{(n)}$ and $T_{n+1} = T_n$. If not, we use Lemma 4.21 with $T = T_n$, $\mathcal{D}_T = \mathcal{D}^{(n)}$, and $t = t_{n+1}$ to obtain $\mathcal{D}^{(n+1)} := \mathcal{D}_{cmb}$ with $\mathcal{D}^{(n+1)} \models_U \mathcal{I}$. We set $T_{n+1} = T_n \cup I_{t_{n+1},s}^{\mathcal{D}^{(n+1)}}$, thus and $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_{n+1}$, as $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_n$ (induction), $Pr_{t_{n+1}}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_{n+1}}^{\mathcal{I}}(\Diamond s)$ (see Lemma 4.21), and $Pr_u^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_{n+1},s}^{\mathcal{D}^{(n+1)}}$ (see Theorem 4.19). □

Thus, to conclude, for any given IDTMC $\mathcal{I}$ and state s, there exists a $\mathcal{D}$ with optimal reachability probability for all states $t \in S^{\mathcal{I}}$

Remark. *Non-attainability of values in EVT intervals.* In general, not every value in $[\underline{\mathbb{E}}[vt_s], \overline{\mathbb{E}}[vt_s]]$ is attainable. In other words, there exists an IDTMC with state s, such that there exists a $x \in [\underline{\mathbb{E}}^{\mathcal{I}}[vt_s], \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]]$ with $\mathbb{E}^{\mathcal{D}}[vt_s] \neq x$ for all $\mathcal{D} \models \mathcal{I}$. This holds even for $\overline{\mathbb{E}}[vt_s] < \infty$.

We provide a short example for this remark.

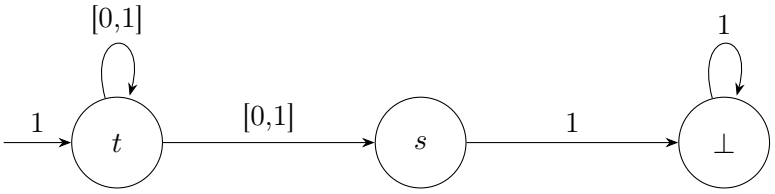


Figure 4.4: An example of a DTMC.

Example 4.23
Consider the IDTMC $\mathcal{I} = (\{t, s, \bot\}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ shown in Fig. 4.4. Clearly, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$ and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 1$. However, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = x$ for any $x \in (0, 1)$, as $\mathbb{E}^{\mathcal{D}}[vt_s] = 1$ if $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ and $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ else.

4.2 Calculating Expected Visiting Times of IDTMCs

We will now have a look at how the *expected visiting times* for IDTMCs can be calculated.

The most intuitive upper and lower bounds for the uEVTs and lEVTs of IDTMCs can be derived by using Eq. (3.2) and the upper and lower transition probability functions and initial distributions defined in Def. 4.1.

Theorem 4.24 (Upper and Lower Bounds for uEVTs and lEVTs)
For any IDTMC $\mathcal{I}$, the uEVT is bounded from above by $\overline{\mathbb{E}}[vt_s] \leq \mathbb{E}_{upp}[vt_s] = \bar{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^\mathsf{T} \cdot \mathbb{E}_{upp}[vt_t]$ and the ℓEVT is bounded from below by $\underline{\mathbb{E}}[vt_s] \geq \mathbb{E}_{low}[vt_s] = \underline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \underline{\mathbf{P}}(t,s)^\mathsf{T} \cdot \mathbb{E}_{low}[vt_t]$ for all $s \in S^{\mathcal{I}}_{tr}$.

Proof 10
For any IDTMC $\mathcal{I}$ there exists a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$, i.e., $\mathcal{D} \models \mathcal{I}$, and $\mathbb{E}^{\mathcal{D}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. As $\mathcal{D} \models \mathcal{I}$, it follows that $\mathbf{P}(s,s') \leq \overline{\mathbf{P}}(s,s')$ for all $s, s' \in S$.

Thus, $\mathbb{E}^{\mathcal{D}}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s)^\mathsf{T} \cdot \mathbb{E}[vt_t] \leq \bar{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^\mathsf{T} \cdot \mathbb{E}_{upp}[vt_t] = \mathbb{E}^{\mathcal{I}}_{upp}[vt_s]$.

The identical argument can be made to prove the lower bound for the ℓEVT. □

As we fix the transition probability functions to either $\underline{\mathbf{P}}$ or $\overline{\mathbf{P}}$, we remove the uncertainty of the IDTMCs and thus obtain a DTMC with a transition matrix that is not row-stochastic. Therefore, we can calculate these bounds by using Eq. (3.2) in PTIME. Moreover, as the transition matrix corresponding to $\underline{\mathbf{P}}$ is row-substochastic, and thus convergent as shown in [KS76], we can use value iteration to compute the lower bounds $\mathbb{E}_{low}$.

However, while these bounds can be computed efficiently, they are not very tight.

Example 4.25
For this take a look at the upper bound of the IDTMC provided in Fig. 4.5. Using Theorem 4.24, we can calculate that $\mathbb{E}_{upp}[vt_{s_4}] = 2$, as $\mathbb{E}_{upp}[vt_{s_1}] = \mathbb{E}_{upp}[vt_{s_2}] = \mathbb{E}_{upp}[vt_{s_3}] = 1$. However, as the IDTMC is acyclic, see Def. 3.11, every state can be visited at most once. Thus, the upper bound $\mathbb{E}_{upp}[vt_{s_4}]$ is clearly not tight. Similarly, $\mathbb{E}_{low}[vt_{s_4}] = 0$, however s_4 needs to be visited at least once. Thus, this lower bound $\mathbb{E}_{low}[vt_{s_4}]$ is not tight either.

In the following, we will therefore take a look at how we can characterize the precise values of the uEVTs and lEVTs.

Theorem 4.26
For a given state s in an IDTMC $\mathcal{I}$, $\overline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{re}$.

Proof 11
By our **??**, there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that s is reachable in $\mathcal{D}$, i.e., $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$. Moreover, as $s \in S^{\mathcal{I}}_{re}$, there exists a DTMC $\mathcal{D}_{re} \models \mathcal{I}$ such that $Pr^{\mathcal{D}_{re}}_s(\bigcirc\Diamond s) = 1$, see Def. 4.14, thus $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Us-

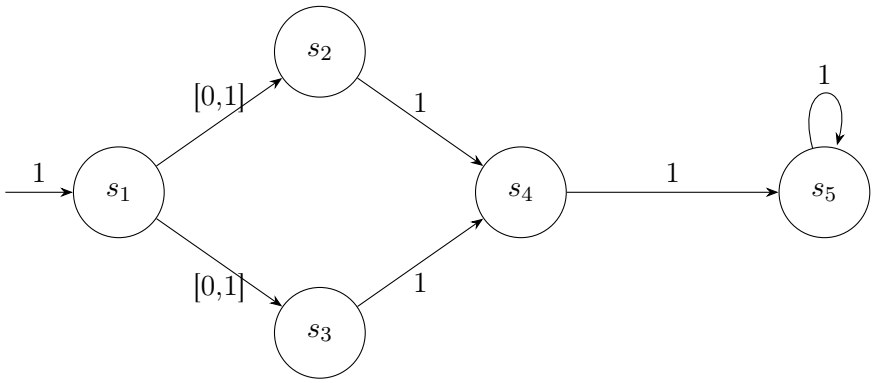


Figure 4.5: An example of a DTMC.

ing Theorem 4.22, there exists a DTMC $\mathcal{D}_{+re} \models \mathcal{I}$ such that $Pr^{\mathcal{D}_{+re}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}_{+re}}_{N(s)}(\Diamond s) = 1$, as $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}_{s}(\bigcirc\Diamond s) = 1$. Therefore, clearly $s \in S^{\mathcal{D}_{+re}}_{re}$, see Def. 3.12 and using Lemma 3.19, we can thus conclude that $\mathbb{E}^{\mathcal{D}_{+re}}[vt_s] = \infty$ and thus $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$.

Conversely, if $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$, then there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = \infty$, i.e. $s \in S^{\mathcal{D}}_{re}$, see Lemma 3.19. Thus, it follows directly that $s \in S^{\mathcal{I}}_{re}$, see Def. 4.14. □

Theorem 4.27
For any $s \in S^{\mathcal{I}}$ in a given IDTMC $\mathcal{I}$, $\underline{\mathbb{E}}[vt_s] = 0$ if and only if $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, i.e., there exists a satisfying DTMC of $\mathcal{I}$, such that s is not reachable.

Proof 12
If $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, then there exists $\mathcal{D} \models \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond s) = 0$, i.e., there exists no path in $\mathcal{D}$ that contains s. As therefore, $vt_s(\pi) = 0$ for all $\pi \in Paths^{\mathcal{D}}_{inf}$, we can conclude, using Def. 3.18, that $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$.

Conversely, if $\underline{Pr}^{\mathcal{I}}(\Diamond s) > 0$, then there exists $\pi \in Paths^{\mathcal{D}}_{inf}$ with $vt_s(\pi) > 0$ for all $\mathcal{D} \models \mathcal{I}$. Therefore, $\mathbb{E}^{\mathcal{D}}[vt_s] > 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] > 0$. □

As shown in [CHK13], computing the maximum or minimum reachability probability is P-complete for the IMDP semantic, and thus also for the UMC semantic.

We will now shortly show that computing ℓEVTs and uEVTs is P-complete, by reducing it to the computation of minimum and maximum reachability probabilities, which is P-complete.

Theorem 4.28
The problem of uEVTs and ℓEVTs is P-complete.

Reduction. Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, we construct the IDTMC $\mathcal{I}_{red} = (S^{\mathcal{I}} \cup \{s, \bot\}, \mathbf{P}^{\mathcal{I}_{red}}, \iota^{\mathcal{I}}_{init})$. We define $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \overline{\mathbf{P}}^{\mathcal{I}}(u,w)$ and $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \underline{\mathbf{P}}^{\mathcal{I}}(u,w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$. Moreover, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = 0$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u,s) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u,s) = 1$ for $u \in T$ and $w \in S^{\mathcal{I}}$. Finally, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s,\bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s,\bot) = 1$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot,\bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot,\bot) = 1$.

Lemma 4.29
The mapping $(\mathcal{I}, T) \mapsto (\mathcal{I}_{red})$ is computable in logarithmic space.

The proof of this lemma will be omitted in this thesis.

Theorem 4.30
Let $\mathcal{D}_{red} \models_U \mathcal{I}_{red}$ denote a DTMC with $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $\mathcal{D}_{max} \models_U \mathcal{I}$ denote a DTMC with $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Then, $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{max}}(\Diamond T)$.

We will prove this theorem by showing that the (maximum) reachability probabilities of T and s are identical in $\mathcal{I}_{red}$ and $\mathcal{D}_{red}$. Then, we will verify that the uEVT of s is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$ and the EVT of s is equal to the reachability probability of s in $\mathcal{D}_{red}$. Lastly, we will show that the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$. By combining these lemmata we will then finally prove the theorem.

Lemma 4.31
$\overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$

Proof 13
As $\mathbf{P}^{\mathcal{D}}(t,s) > 0$ if and only if $t \in T$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, it follows from Lemma 3.8 that $Pr^{\mathcal{D}}(\Diamond s) = \sum_{t \in T} \mathbf{P}(t,s) \cdot Pr^{\mathcal{D}}(\Diamond T)$. Because, $\mathbf{P}^{\mathcal{D}}(t,s) = 1$ for all $t \in T$, we can conclude that $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond s)$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$. □

Lemma 4.32
$\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{red}}(\Diamond s)$

Proof 14
$\mathbb{E}^{\mathcal{D}}[vt_s] = Pr(\Diamond s) \cdot \mathbb{E}^{\mathcal{D}}_s[vt_s]$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, see Lemma 3.22. As, $\overline{Pr}^{\mathcal{I}_{red}}(s,t) > 0$ if and only if $t = \tau$ and τ is absorbing, it follows that $\mathbb{E}^{\mathcal{D}}_s[vt_s] = 1$, as $Pr^{\mathcal{D}}_s(\bigcirc \Diamond s) = 0$, which concludes the proof. □

Lemma 4.33
$Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{max}}(\Diamond T)$

Proof 15
Firstly, if $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ (as defined in the construction), then $Pr^{\mathcal{D}_{red}}(\Diamond s) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$, see Lemma 4.32. Moreover, using Lemma 4.31, we can then conclude that $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$.

As $Pr^{\mathcal{D}_{red}}(\Diamond T)$ and $Pr^{\mathcal{D}_{max}}(\Diamond T)$ are only dependent on $\mathbf{P}^{\mathcal{D}_{red}}(u, w)$ and $\mathbf{P}^{\mathcal{D}}(u, w)$ with $u \in S^{\mathcal{I}} \backslash T$ respectively, see Lemma 3.8, and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \overline{\mathbf{P}}^{\mathcal{I}}(u, w)$ as well as $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}}(u, w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$, we can conclude that $\overline{Pr}^{\mathcal{I}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$. By the definition of our construction, $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$ (see first paragraph of this proof) which concludes the proof. □

We can now prove Theorem 4.30.

Proof 16
We simply need to combine the results of the lemmata. According to Lemma 4.33 $Pr^{\mathcal{D}_{max}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond T)$. Furthermore, using Lemma 4.31, $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$. Finally, Lemma 4.32 shows that $Pr^{\mathcal{D}_{red}}(\Diamond s) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. Thus, combining these results $Pr^{\mathcal{D}_{max}}(\Diamond T) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. By definition $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Therefore, the given maximum reachability probability $\overline{Pr}^{\mathcal{I}}(\Diamond T)$ of any IDTMC $\mathcal{I}$ can be determined through calculating uEVTs.□

4.2.1 UMC semantic

In this section we examine how the precise values of the uEVTs and lEVTs can be calculated for IDTMCs under the UMC semantic.

As defined, there can be multiple DTMCs $\mathcal{D}_i$ that satisfy an IDTMC $\mathcal{I}$ with differing transition probability functions $\mathbf{P}^{\mathcal{D}_i}$ and initial distributions $\iota_{init}^{\mathcal{D}_i}$. To determine the uEVT and ℓEVT for a given state $s \in S^{\mathcal{I}}$, we thus have to find the satisfying DTMCs that maximize and minimze Eq. (3.2), respectively. In other words, we need to solve the equations

$$\mathbb{E}[vt_s] = \text{opt}_{\mathcal{D} \models \mathcal{I}} \left(\iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}^{\mathcal{D}}(t, s) \cdot \mathbb{E}[vt_t] \right), \qquad \text{opt} \in \{\min, \max\}.$$

In [CHK13], the authors construct the following NLP (which we have rewritten slightly) to compute any maximum or minimum reachability, respectively, for a set of nodes $T \subseteq S$, using the fixed-initial-state-definition.

$$
\begin{aligned}
&\text{opt} && \sum_{s\in S\setminus T} x_s && \text{opt} \in \{\max, \min\} && \\
&\text{s.t.} && x_s \sim \sum_{s'\in S\setminus T} \mathbf{P}(s,s')x_{s'} + \sum_{s'\in T} \mathbf{P}(s,s') && \forall s \in S\setminus T \text{ and } \sim\in\{\leq,\geq\} && (1)\\
& && x_s = 1 && \forall s \in T && (2)\\
& && \sum_{s_j\in N(s)} \mathbf{P}(s_i,s_j) = 1 && \forall s_i \in S_{tr} && (3)\\
& && \underline{\mathbf{P}}^{\mathcal{I}}(s_i,s_j) \leq \mathbf{P}(s_i,s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i,s_j) && \forall s_i,s_j \in S^{\mathcal{I}}_{tr} && (4)
\end{aligned}
$$

(Allowed only max + $\leq$ and min + $\geq$)

As shown in [CHK13], this NLP can be solved in PTIME. Thus, we can calculate the maximum or minimum reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ or $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ of a state s in polynomial time. As we showed in Theorem 4.22 in Sec. 4.1.1, there exist a DTMCs $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ for any given IDTMC $\mathcal{I}$ that have maximum or minimum reachability from all states to s, i.e., $Pr_t^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ or $Pr_t^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all $t \in S^{\mathcal{I}}$. Thus, clearly also $Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ or $Pr^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $Pr^{\mathcal{D}_{min}}(\bigcirc\Diamond s) = \underline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$. Using Eq. (3.6), we can then calculate $\mathbb{E}^{\mathcal{D}_{max}}[vt_s]$ or $\mathbb{E}^{\mathcal{D}_{min}}[vt_s]$.

Theorem 4.34
Given $\mathcal{D}_{max} \models \mathcal{I}$, there exists no $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$. Conversely, given $\mathcal{D}_{min}$, there exists no $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] > \mathbb{E}^{\mathcal{D}}[vt_s]$. In other words $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$.

We will prove this theorem by contradiction.

Proof 17
Let us assume that there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$. Using Eq. (3.6), it thus follows that either $Pr^{\mathcal{D}}(\Diamond s) > Pr^{\mathcal{D}_{max}}(\Diamond s)$ or $Pr^{\mathcal{D}}(\bigcirc\Diamond s) > Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s)$. However, as $Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ this leads to a contradiciton. The proof for $\mathcal{D}_{min}$ follows analogously by reversing the inequalities. □

In other words, we can calculate the ℓEVT and uEVT of a state s in a given IDTMC under the UMC semantic by determining the maximum reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ or the minimum reachability probabilities $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$.

4.2.2 IMDP semantic

In this subsection we will look at how uEVTs and lEVTs can be calculated for the IMDP semantic.

Recall, that we concluded in Sec. 4.1.1 that the uEVT and ℓEVT of a given IDTMC $\mathcal{I}$ can be calculated by maximizing or minimizing the term $\sum_{t \in S_{tr} \cap in} \frac{Pr_t(\Diamond s)}{1 - Pr_s(\bigcirc \Diamond s)} \cdot \iota_{init}(t)$ for all DTMCs $\mathcal{D} \models \mathcal{I}$. Intuitively, we can thus find the lEVTs and uEVTs by maximizing or minimizing the reachability probability from the initial state t to the given state s ($Pr_t(\Diamond s)$) and from s back to itself ($Pr_s(\bigcirc \Diamond s)$). As shown in [CHK13] the problem of finding this supremum or infimum for a given IDTMC is P-complete. Thus, we can derive the two suprema or infima belonging to (at most) two different satisfying DTMCs $\mathcal{D}_{s_{in} \to s}$ and $\mathcal{D}_{s \to s}$ ($\mathcal{D}_{s_{in} \to s},\ \mathcal{D}_{s \to s} \models_I \mathcal{I}$).

As we have defined the IMDP semantic using adversarial behaviour (?), notice that we can define a policy that choses the probability distribution specified by $\mathcal{D}_{s_{in} \to s}$ until state s is visited for the first time and then choses the probability distribution specified by $\mathcal{D}_{s \to s}$. This policy can be characterized as having two stages where we transition from the first to the second stage when passing through s for the first time. Notice however, that the stochastic process of the Markov chain/MDP (?) would no longer be memoryless, i.e. the Markov Property (SEE ...) would break, as the process is not only dependent on the current state but also on what stage the policy is in. To solve this problem, we propose a remapping/reduction (state augmentation).

Construction. Given an IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, the previously defined $\mathcal{D}_{s_{in} \to s}$, $\mathcal{D}_{s \to s}$ (corresponding to either the suprema or infima) and a state s (for which we want to calculate the EVT), we define $\mathcal{D}^s_{aug}$, with $S^{\mathcal{D}^s_{aug}} = S_{st1} \cup S_{st2}$, where the state set for stage 1 is $S_{st1} = \{s^{(1)} \mid s \in S^{\mathcal{I}}_{tr}\}$ (transient states in $\mathcal{I}$) and the state set for stage 2 is $S_{st2} = \{s^{(2)} \mid s \in S^{\mathcal{I}}\}$. Furthermore, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in} \to s}}(s_1, s_2)$ for all $s_1^{(1)} \in S_{st1}$, $s_2^{(1)} \in S_{st1} \backslash \{s\}$ and $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(2)}) = \mathbf{P}^{\mathcal{D}_{s_{in} \to s}}(s_1, s_2)$ for $s_1^{(1)} \in S_{st1}$, $s_2^{(2)} \in S_{st2} \cap S^{\mathcal{I}}_{re}$. Similarly, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(2)}, s_2^{(2)}) = \mathbf{P}^{\mathcal{D}_{s \to s}}(s_1, s_2)$ for all $s_1^{(2)}, s_2^{(2)} \in S_{st2}$. Moreover, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s_i^{(2)}) = \mathbf{P}^{\mathcal{D}_{s \to s}}(s, s_i)$ for $s^{(1)} \in S_{st1}$ and $s_i \in N^{\mathcal{I}}(s)$. Lastly, $\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(1)}) = \iota^{\mathcal{D}_{s_{in} \to s}}_{init}(t)$ for all $t^{(1)} \in S_{st1}$ and $\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(2)}) = \iota^{\mathcal{D}_{s_{in} \to s}}_{init}(t)$ for all $t^{(2)} \in S_{st2} \cap S_{re}$, while $\iota^{\mathcal{D}^s_{aug}}_{init}(t^{(2)}) = 0$ for all $t^{(2)} \in S_{st2} \backslash S_{re}$.

Example 4.35
Recall the IDTMC $\mathcal{I}$ in Fig. 4.1, displayed again in Fig. 4.6. Fig. 4.7 shows $\mathcal{D}^{s_4}_{aug}$, i.e., the construction of the maximizing DTMC for the EVT of $\mathcal{I}$.

(We will now show that this remapping / characterization of this history-dependent policy is coincides (?) with Def. 4.8.)

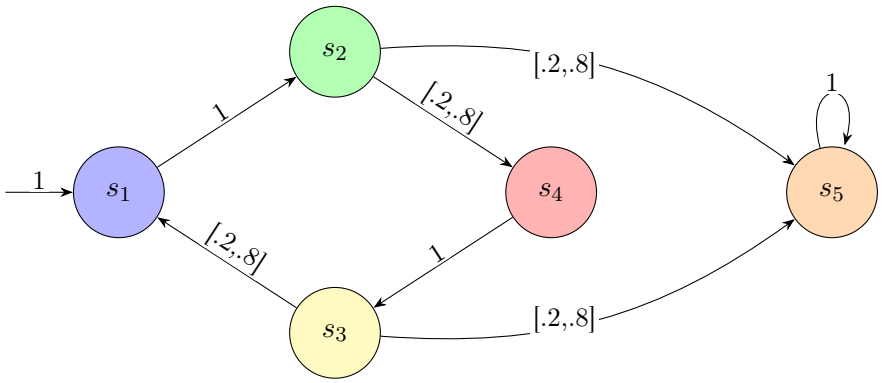


Figure 4.6: target state in red

Lemma 4.36
Given and IDTMC $\mathcal{I}$ and the corresponding $\mathcal{D}^s_{aug}$, it holds that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$

Proof 18
To prove this lemma, we will show that Def. 4.8 holds for $\mathcal{D}^s_{aug}$. For this, we will first define a mapping $\tau : S^{\mathcal{D}^s_{aug}} \to S^{\mathcal{I}}$. For this, we need to account for S_{st1} and S_{st2}. We define $\tau(s_i^{(1)}) = s_i$ for all $s_i^{(1)} \in S_{st1} \cap S_{tr}$ and $\tau(s_i^{(2)}) = s$ for all $s_i^{(2)} \in S_{st2}$. Thus, for all $s \in S^{\mathcal{I}}_{tr}$, there exists precisely one $s^{(1)} \in S_{st1}$ with $\tau(s^{(1)}) = s$ and $\iota^{\mathcal{D}_{aug}}_{init}(s^{(1)}) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$ and $s^{(2)} \in S_{st2}$ with $\tau(s^{(2)}) = 0$. Moreover, for all $s \in S^{\mathcal{I}}_{re}$, there exists precisely one $s^{(2)} \in S_{st2}$ with $\tau(s^{(2)}) = s$ and $\iota^{\mathcal{D}_{aug}}_{init}(s^{(2)}) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$ and $s^{(1)} \notin S_{st1}$. Therefore, we can conclude that condition (i) of Def. 4.8 holds.

Secondly, condition (ii) clearly holds for all $s_i^{(2)} \in S_{st2}$, as all of its transition are defined to be identical to that of $\mathcal{D}_{s\to s}$ and $\mathcal{D}_{s\to s} \models \mathcal{I}$. Thus, it remains only to show that condition (ii) also holds for all states in S_{st1}. For all states $s_1^{(1)} \in S_{st1}\backslash\{s\}$, $\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_1, s_2)$, for all $s_2 \in S^{\mathcal{D}_{s_{in}\to s}}_{tr}$ and $s_2 \in S^{\mathcal{D}_{s_{in}\to s}}_{re}$. Moreover, for $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s_i^{(1)}) = 0$ and $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s_i^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s, s_i)$. As, $\mathcal{D}_{s_{in}\to s}$, $\mathcal{D}_{s\to s} \models \mathcal{I}$, we can thus conclude that condition (ii) also holds for all states in S_{st1}. As both conditions of Def. 4.8 are fulfilled, we can thus conclude that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$. □

Construction. For a given DTMC $\mathcal{D}$ and a state $s \in S$, consider the DTMC $\mathcal{D}_{abs} = (S^{\mathcal{D}_{abs}}, \mathbf{P}^{\mathcal{D}_{abs}}, \iota^{\mathcal{D}}_{init})$ with the set of states $S^{\mathcal{D}_{abs}} = S^{\mathcal{D}}_{tr} \cup \{s_{re}\}$. Moreover, the probability function $\mathbf{P}_{abs}$ is identical to $\mathbf{P}^{\mathcal{D}}$, except the transitions $\mathbf{P}_{abs}(s, s') = 0$ for all $s' \in S$ and $\mathbf{P}_{abs}(s, s) = 1$, as well as $\mathbf{P}^{\mathcal{D}}(s_1, s_2) = \mathbf{P}^{\mathcal{D}_{abs}}(s_1, s_{re})$, for all $s_1 \in S^{\mathcal{D}}_{tr}$, $s_2 \in S^{\mathcal{D}}_{re}$ and further $\mathbf{P}^{\mathcal{D}_{abs}}(s_{re}, s_{re}) = 1$. In other words, $\mathcal{D}_{abs}$ is modified only in such a way that s is absorbing and all recurrent states are summarized into one state.

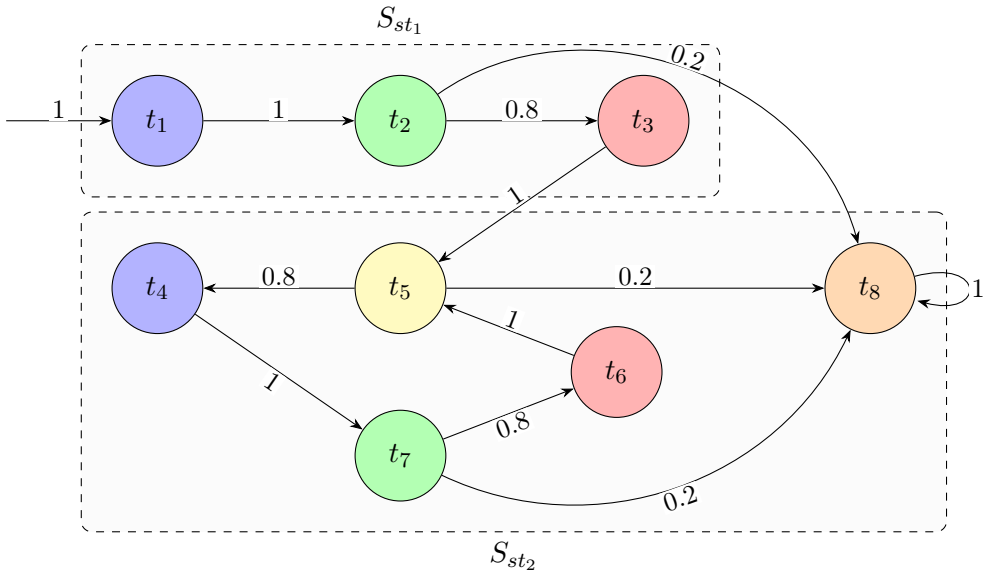


Figure 4.7: $\tau(t_1) = \tau(t_4) = s_1$, $\tau(t_2) = \tau(t_7) = s_2$, $\tau(t_5) = s_3$, $\tau(t_3) = \tau(t_6) = s_4$. Target state in red.

Lemma 4.37
The reachability probabilities $Pr^{\mathcal{D}}(\Diamond s) = Pr^{\mathcal{D}_{abs}}(\Diamond s)$ are equal.

Proof 19
If the set of initial states S_{init} includes s then trivially the reachability probability is 1 for both $\mathcal{D}$ and $\mathcal{D}_{abs}$.

As $Pr(\Diamond s) = Pr(S \backslash \{s\} \mathsf{U} s)$ (until operator), we can reason/conclude using Def. 3.3 and Def. 3.5 that $Pr(\Diamond s) = \iota^{\mathcal{D}}_{init}(\pi[0]) \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1])$. As there is no factor $\mathbf{P}(s, s')$ for any $s' \in S$ in this product, as well as no factor of the form $\mathbf{P}(s_1, s_2)$, where $s_1 \in S^{\mathcal{D}}_{re}$ and the probabilty functions $\mathbf{P}$ and $\mathbf{P}_{abs}$ are identical for all transitions except those, we can conclude that Lemma 4.37 is true. □

Thus, we can reason that transitions from the state s itself and recurrent states have no influence on the reachability probability $Pr^{\mathcal{D}}(\Diamond s)$.

Lemma 4.38
Let $Pr^{max}(\Diamond s)$ and $Pr^{max}_s(\bigcirc \Diamond s)$ be the maximum reachability probabilities of $\mathcal{I}$. It holds that $Pr^{D^s_{aug}}(\Diamond s) = Pr^{max}(\Diamond s)$ and $Pr_s^{D^s_{aug}}(\bigcirc \Diamond s) = Pr^{max}_s(\bigcirc \Diamond s)$

Proof 20
First, we will show that $Pr^{D^s_{aug}}(\Diamond s) = Pr^{max}(\Diamond s)$. Notice, that for all $s \in S_{st1} \backslash \{s\}$, the probability function $\mathbf{P}^{\mathcal{D}^s_{aug}}$ is identical to $\mathbf{P}^{\mathcal{I}}$, i.e. it is identical for all transient states except s. Using Lemma 4.37, we can thus conclude that $Pr^{D^s_{aug}}(\Diamond s) = Pr^{max}(\Diamond s)$.

Next, we will show that $Pr_s^{D^s_{aug}}(\bigcirc\Diamond s) = Pr_s^{max}(\bigcirc\Diamond s)$. Trivial because (basicaly/literaly) the same DTMC. □

4.2.3 Special Case: Acyclic IDTMCs

Definition 4.39 (Acyclic IDTMC)
An IDTMC $\mathcal{I}$ is acyclic if and only if all satisfying DTMCs $\mathcal{D}$ ($\mathcal{D} \models \mathcal{I}$) are acyclic.

– l/uEVT is simply maximum/minimum reachability (which can be calc in P, cite paper).

4.2.4 Value Iteration

Explain why useful & interesting to define Bellman operator usw. (Runtime advantages, etc.)

In [KTDB19], the authors prove

4.3 Exkurs: aIMCs

Definition 4.40 (Augmented Interval Markov Chains)
An augmented Interval Markov Chain (aIMC) $\mathcal{A}$ is a tuple $(S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init}, C)$, where $(S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ is an IDTMC $\mathcal{I}$ and $C \subseteq S^4$ is a set of *edge equality constraints*. Similar to the once-and-for-all semantic for IDTMCs, see **??**, we define, that an arbitrary DTMC $\mathcal{D}$ satisfies an aIMC $\mathcal{A}$ if it satisfies the corresponding IDTMC $\mathcal{I}$ and $\mathbf{P}^{\mathcal{D}}(s_1, s_2) = \mathbf{P}^{\mathcal{D}}(s_3, s_4)$ for all $(s_1, s_2, s_3, s_4) \in C$. [Cho17]

In other words, aIMCs extend the definition of IMCs with an "edge-constraint" which enforces certain transition probabilities to be equal. A restriction of the quantitative reachability problem, which we discussed in Sec. 3.3 and Sec. 4.1.1, is the qualitative reachability problem, where is restricted to $p \in \{0, 1\}$ (see Sec. 4.1.1).

In [Cho17], the authors prove that the qualtitative reachability problem for aIMCs is NP-complete. Thus, as trivially the qualitative problem can be reduced to the quantitative problem, we can thus deduce that the quantitative problem for aIMCs is NP-hard. As we have shown in ...

4.4 Stationary Distributions in IDTMCs

As stated in **??** the k-th power of the transition matrik $\mathcal{T}^k$ contains the probabilities of transitioning from any state i to any other state j in precisely k steps. Thus, the product $\iota_{init}^{\mathsf{T}} \cdot \mathcal{T}^k$ characterizes the probability distribution of the current state of the given DTMC in step k. We define this product as $\theta_k^{\mathsf{T}} = \iota_{init}^{\mathsf{T}} \cdot \mathcal{T}^k$, where the i-th entry of θ_k characterizes the probability that the current state in step k is i. We can also define θ_k iteratively through $\theta_k^{\mathsf{T}} = \theta_{k-1}^{\mathsf{T}} \cdot \mathcal{T}$

Definition 4.41 (Stationary Distributions in DTMCs)
A distribution $\theta_k^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is stationary if and only if $\theta_k^{\mathsf{T}} = \theta_k^{\mathsf{T}} \cdot \mathcal{T}^{\mathcal{D}}$, i.e. the probability distribution of the current state remains unchanged through future steps. We denote the stationary distribution as $\theta^{\mathcal{D}}$, where $\theta^{\mathcal{D}}(s)$ denotes the entry of $\theta^{\mathcal{D}}$ which corresponds to state s.

Definition 4.42 (Stationary Distributions in IDTMCs)
For a given IDTMC $\mathcal{I}$, we define $\Theta = \{\theta^{\mathcal{D}} \mid \mathcal{D} \models \mathcal{I}\}$ as the set of stationary distributions. Furthermore, for a given state $s \in S^{\mathcal{I}}$, we define $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ such that $\underline{\theta}^{\mathcal{I}}(s) = \inf_{\mathcal{D} \models \mathcal{I}} \theta^{\mathcal{D}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s) = \sup_{\mathcal{D} \models \mathcal{I}} \theta^{\mathcal{D}}(s)$.

In contrast to EVTs, when considering the stationary distribution of a DTMC or IDTMC, we are interested in the recurrent, not transient, states.

Theorem 4.43
If θ is stationary, then $\theta(i) = 0$ if the corresponding state s_i is transient and $\theta(i) > 0$ if s_i is recurrent.

We provide a short proof.

Proof 21
Assume that some s is transient and for the stationary distribution $\theta(s) > 0$. As θ is stationary, there exists some $k \in \mathbb{N}_{>0}$ such that $\theta_{k'}$ is stationary for $k' \geq k$. Thus, there is a nonzero probability that the current state in step $k' \geq k$ is s. However, as s is transient it follows that $Pr_s(\bigcirc\Diamond s) < 1$, see Def. 3.12. Therefore, there must be some $k'' > k$ such that $\theta_{k''}(i) = 0$. This contradicts the stationary property of θ. Now, assume that some s is recurrent and for the stationary distribution $\theta(s) = 0$. As s is reachable by definition/construction, see **??**, we can conclude that $Pr(\Diamond s) > 0$. Thus, there exists some $k' \in \mathbb{N}_{>0}$ such that $\theta_{k'}(i) > 0$. As s_i is recurrent, it follows that $Pr_{s_i}(\bigcirc\Diamond s_i) = 1$, see again Def. 3.12. Thus, there exist infinitely many $k'' \in \mathbb{N}_{>0}$ with $k'' > k'$ and $\theta_{k''}(i) > 0$. Therefore, θ_k is not stationary. □

Notice, that any maximal set of connected recurrent states, i.e., every state is connected to some other state in the set, in a DTMC, is a BSCC, see Def. 3.10. Firstly, a set of

connected recurrent states in a DTMC is strongly connected. By definition, see **??**, all of the recurrent states in the set are reachable. As the return probability $Pr_{s_{re}}(\bigcirc\Diamond S_{re})$ is equal to 1 for any recurrent state s_re, see Def. 3.12, any state that is a successor to s_{re} must also be a predecessor, i.e., every edge in this set is bidirectional (in a sense). Thus, trivially strongly connected.

→ It is SCC

Clearly also BSCC

(Bridge: recurrent groups are BSCC & BSCC is irreducible DTMC (perhaps prove this shortly)) We denote stationary distribution of a DTMC $\mathcal{D}$ restricted to a BSCC B as $\theta^{\mathcal{D}|_B}$.

As proven in [Kul16], the stationary distribution $\theta^{\mathcal{D}}$ of the recurrent states of a DTMC $\mathcal{D}$ is characterized by

$$\theta^{\mathcal{D}}(s) = Pr^{\mathcal{D}}(\Diamond B) \cdot \theta^{\mathcal{D}|_B}(s) \ , \tag{4.2}$$

for all $s \in B$. Moreover, in [MKQW25], it is shown that the stationary distribution within a BSCC can be computed through its EVTs. For this they construct a DTMC $\mathcal{B}\,\text{↱}^{\hat{v}}$ for every BSCC B with any arbitrary state $v \in S^{\mathcal{D}|_B}$ in which all incoming transitions of v are redirected to the fresh absorbing state $\hat{v}$.

Construction. More precisely, $\mathcal{B}\,\text{↱}^{\hat{v}} = (S^{\mathcal{D}|_B} \cup \{\hat{v}\}, \mathbf{P}^{\mathcal{B}\text{↱}^{\hat{v}}}, \iota_{init}^{\mathcal{B}\text{↱}^{\hat{v}}})$, where $\mathbf{P}^{\mathcal{B}\text{↱}^{\hat{v}}}(\hat{v},\hat{v}) = 1$, $\mathbf{P}^{\mathcal{B}\text{↱}^{\hat{v}}}(s,\hat{v}) = \mathbf{P}(s,v)$ and $\mathbf{P}^{\mathcal{B}\text{↱}^{\hat{v}}}(s,v) = 0$ for all $s \in S^{\mathcal{D}|_B}$, $\mathbf{P}^{\mathcal{B}\text{↱}^{\hat{v}}}(s,t) = \mathbf{P}(s,t)$ for all $s \in S$, $t \in S\backslash\{v\}$, and $\iota_{init}^{\mathcal{B}\text{↱}^{\hat{v}}}(v) = 1$.

The stationary distribution within a DTMC is then given by

$$\theta^{\mathcal{D}|_B}(s) = \frac{\mathbb{E}^{\mathcal{B}\text{↱}^{\hat{v}}}[vt_s]}{\sum_{t\in S}\mathbb{E}^{\mathcal{B}\text{↱}^{\hat{v}}}[vt_t]} \ , \tag{4.3}$$

as proven in [MKQW25].

Assumption 2
In the rest of this section, we assume that all IDTMCs are graph preserving, meaning that there exist no lower transition probability function such that $\underline{\mathbf{P}}^{\mathcal{I}}(s,t) = 0$ for any $s,t \in S^{\mathcal{I}}$.

Notice, that all DTMCs $\mathcal{D}_i$ (for UMC semantic) satisfying an IDTMC $\mathcal{D}_i \models \mathcal{I}$ have the same edge relations, i.e., if $\mathbf{P}^{\mathcal{D}_j}(u,w) > 0$ for some $u,w \in S^{\mathcal{I}}$, then $\mathbf{P}^{\mathcal{D}_k}(u,w) > 0$ for all $\mathcal{D}_k$. (schlechte notation; auch wieder: eig u,w in $S^{\mathcal{D}}$, bzw. die states in $\mathcal{D}$, die auf gewieses u in IMC verweisen).

As all IDTMCs are graph preserving, we can conclude that for a given IDTMC $\mathcal{I}$, any state $s \in S^{\mathcal{I}}_{re}$ and BSCC B, if $s \in B$ then $s \notin B'$ for all BSCC $B' \neq B$. In other words, every state $S^{\mathcal{I}}_{re}$ corresponds to precisely one BSCC in $\mathcal{I}$.

Proof 22
Assume that for some IDTMC $\mathcal{I}$ there exists a state s and two BSCCs $B^{\mathcal{D}}, B'^{\mathcal{D}'}$ with $B^{\mathcal{D}} \neq B'^{\mathcal{D}'}$ and $\mathcal{D}, \mathcal{D}' \models \mathcal{I}$ such that $s \in B$ and $s \in B'$. As $B \neq B'$, there exists some $t \in (B \backslash B') \cup (B' \backslash B)$, i.e., there exists some $t \in S^{\mathcal{I}}$ only in B or B'. Let us assume $t \in B^{\mathcal{D}}$. (ist das ok so, halt um triviale Fallunterscheidung zu vermeiden) Thus by definition, see Def. 3.10, there exists a path from s to t in $\mathcal{D}$. However, as noted the graph relations for all satisfying DTMCs of graph preserving IDTMCs are identical. Thus, there must also exist a path from s to t in $\mathcal{D}'$. However, $t \notin B'^{\mathcal{D}'}$, which contradicts Def. 3.10. □

Thus, we can conclude that for graph preserving IDTMCs that any state $s \in S^{\mathcal{I}}$ belongs only to a singular BSCC, i.e., the BSCCs for all satisfying DTMCs are identical.

Therefore, we can calculate the stationary distributions

$$\underline{\theta}^{\mathcal{I}}(s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \cdot \frac{\underline{\mathbb{E}}^{\mathcal{B}^{\lceil \hat{v}}}[vt_s]}{\sum_{t \in S} \mathbb{E}^{\mathcal{B}^{\lceil \hat{v}}}[vt_t]} \ , \tag{4.4}$$

similar to Eq. (4.2) using Eq. (4.3).

Show: when $\mathbb{E}[vt_s]$ minimal then sum of $\mathbb{E}[vt_t]$ is maximal (and other way around). Bzw die summe aller evts ist für alle satisfying DTMCs gleich (–> nicht wahr; vllt auch ein interessantes Ergebnis??)

This directly concludes the optimality of the BSCC restricted stat. distribution.

(Trivially then uppper bound for stationary distribution given by uEVT and lower bound given by ℓEVT)

Chapter 5

Main Contribution

Chapter 6

Conclusion

Appendix A

Appendix

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Bibliography

[ADD00] Robert B. Ash and Catherine C. Doleans-Dade. *Probability and Measure Theory*. Elsevier Science, 2000.

[Afr20] Feri Afrinaldi. Exploring product lifecycle using Markov chain. *Procedia Manufacturing*, 43:391–398, 2020. Sustainable Manufacturing - Hand in Hand to Sustainability on Globe: Proceedings of the 17th Global Conference on Sustainable Manufacturing.

[AGM95] Samson Abramsky, Dov M. Gabbay, and Thomas S. E. Maibaum, editors. *Handbook of Logic in Computer Science*, volume 3. Oxford University Press, USA, 1995.

[AN14] Subil Abraham and Suku Nair. Cyber Security Analytics: A Stochastic Model for Security Quantification Using Absorbing Markov Chains. *Journal of Communications*, pages 899–907, 12 2014.

[Bad] Thom Badings. Robust Verification of Stochastic Systems: Guarantees in the Presence of Uncertainty.

[BDL+17] Anicet Bart, Benoit Delahaye, Didier Lime, Eric Monfroy, and Charlotte Truchet. Reachability in Parametric Interval Markov Chains using Constraints, 2017.

[BDLM21] Giovanni Bacci, Benoît Delahaye, Kim G. Larsen, and Anders Mariegaard. *Quantitative Analysis of Interval Markov Chains*, pages 57–77. Springer International Publishing, Cham, 2021.

[BK08] Christal Baier and {Joost P.} Katoen. *Principles of Model Checking*. MIT Press, United States, May 2008.

[BKL+17] Christel Baier, Joachim Klein, Linda Leuschner, David Parker, and Sascha Wunderlich. Ensuring the Reliability of Your Model Checker: Interval Iteration for Markov Decision Processes. In Rupak Majumdar and Viktor Kunčak, editors, *Computer Aided Verification*, pages 160–180, Cham, 2017. Springer International Publishing.

[BLNW14] Fabrizio Biondi, Axel Legay, Bo Friis Nielsen, and Andrzej Wąsowski. Maximizing entropy over Markov processes. *Journal of Logical and Algebraic Methods in Programming*, 83(5):384–399, 2014. The 24th Nordic Workshop on Programming Theory (NWPT 2012).

[BLW13] Michael Benedikt, Rastislav Lenhardt, and James Worrell. LTL Model Checking of Interval Markov Chains. In Nir Piterman and Scott A. Smolka, editors, *Tools and Algorithms for the Construction and Analysis of Systems*, pages 32–46, Berlin, Heidelberg, 2013. Springer Berlin Heidelberg.

[Cas09] Hal Caswell. Stage, age and individual stochasticity in demography. *Oikos*, 118(12):1763–1782, 2009.

[CHK13] Taolue Chen, Tingting Han, and Marta Kwiatkowska. On the complexity of model checking interval-valued discrete time Markov chains. *Information Processing Letters*, 113(7):210–216, 2013.

[Cho17] Ventsislav Chonev. Reachability in Augmented Interval Markov Chains, 2017.

[CSH08] Krishnendu Chatterjee, Koushik Sen, and Thomas A. Henzinger. Model-Checking ω-Regular Properties of Interval Markov Chains. In Roberto Amadio, editor, *Foundations of Software Science and Computational Structures*, pages 302–317, Berlin, Heidelberg, 2008. Springer Berlin Heidelberg.

[FJ90] Noah E. Friedkin and Eugene C. Johnsen. Social influence and opinions. *The Journal of Mathematical Sociology*, 15(3-4):193–206, 1990.

[FRG13] Lance Fiondella, Sanguthevar Rajasekaran, and Swapna S. Gokhale. Efficient Software Reliability Analysis With Correlated Component Failures. *IEEE Transactions on Reliability*, 62(1):244–255, 2013.

[Gal17] Serge Galam. The Trump phenomenon: An explanation from sociophysics. *International Journal of Modern Physics B*, 31(10):1742015, April 2017.

[Gok04] Swapna S. Gokhale. Quantifying the Variance in Application Reliability. In *Proceedings of the 10th IEEE Pacific Rim International Symposium on Dependable Computing (PRDC'04)*, PRDC '04, page 113–121, USA, 2004. IEEE Computer Society.

[GT02] S.S. Gokhale and K.S. Trivedi. Reliability prediction and sensitivity analysis based on software architecture. In *13th International Symposium on Software Reliability Engineering, 2002. Proceedings.*, pages 64–75, 2002.

[GWHT04] Swapna S. Gokhale, W. Eric Wong, J.R. Horgan, and Kishor S. Trivedi. An analytical approach to architecture-based software performance and reliability prediction. *Performance Evaluation*, 58(4):391–412, 2004.

[HBK17] Lisa Hutschenreiter, Christel Baier, and Joachim Klein. Parametric Markov Chains: PCTL Complexity and Fraction-free Gaussian Elimination. *Electronic Proceedings in Theoretical Computer Science*, 256:16–30, 2017.

[JJK22] Nils Jansen, Sebastian Junges, and Joost-Pieter Katoen. Parameter Synthesis in Markov Models: A Gentle Survey, 2022.

[JL91] B. Jonsson and K.G. Larsen. Specification and refinement of probabilistic processes. In *[1991] Proceedings Sixth Annual IEEE Symposium on Logic in Computer Science*, pages 266–277, 1991.

[Kat16] Joost-Pieter Katoen. The Probabilistic Model Checking Landscape. In *Proceedings of the 31st Annual ACM/IEEE Symposium on Logic in Computer Science*, LICS '16, page 31–45, New York, NY, USA, 2016. Association for Computing Machinery.

[KS76] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Undergraduate Texts in Mathematics. Springer, 1976.

[KTDB19] Thomas Krak, Natan T'Joens, and Jasper De Bock. Hitting Times and Probabilities for Imprecise Markov Chains. In Jasper De Bock, Cassio P. de Campos, Gert de Cooman, Erik Quaeghebeur, and Gregory Wheeler, editors, *Proceedings of the Eleventh International Symposium on Imprecise Probabilities: Theories and Applications*, volume 103 of *Proceedings of Machine Learning Research*, pages 265–275. PMLR, 03–06 Jul 2019.

[KU02] Igor O. Kozine and Lev V. Utkin. Interval-Valued Finite Markov Chains. *Reliable Computing*, 8(2):97–113, 2002.

[Kul16] Vidyadhar Kulkarni. *Modeling and analysis of stochastic systems: Third edition*. 11 2016.

[Kuz91] V. P. Kuznetsov. *Interval Statistical Models*. Radio and Communication, 1991.

[LY08] David G. Luenberger and Yinyu Ye. *Linear and Nonlinear Programming*, volume 228 of *International Series in Operations Research & Management Science*. Springer, 3rd edition, 2008.

[MAK06] Yasunari Matsuno, Yoshihiro Adachi, and Yasushi Kondo. Application of Markov Chain Model to Calculate the Average Number of Times of Use of a Material in Society. An Allocation Methodology for Open-Loop Recycling. Part 1: Methodology Development. *The International Journal of Life Cycle Assessment*, 11(5):354–360, 2006.

[Mar06] Andrey Andreyevich Markov. Extension of the law of large numbers to dependent quantities. *Izv. Fiz.-Matem. Obsch. Kazan Univ.(2nd Ser)*, 15(1):135–156, 1906.

[MKQW25] Hannah Mertens, Joost-Pieter Katoen, Tim Quatmann, and Tobias Winkler. Computing Expected Visiting Times and Stationary Distributions in Markov Chains: Fast and Accurate. *Journal of Automated Reasoning*, 69(3), August 2025.

[MMAG11] Indika Meedeniya, Irene Moser, Aldeida Aleti, and Lars Grunske. Architecture-based reliability evaluation under uncertainty. In *Proceedings of the Joint ACM SIGSOFT Conference – QoSA and ACM SIGSOFT Symposium – ISARCS on Quality of Software Architectures – QoSA and Architecting Critical Systems – ISARCS*, QoSA-ISARCS '11, page 85–94, New York, NY, USA, 2011. Association for Computing Machinery.

[Neu77] C. Neumann. *Untersuchungen über das logarithmische und Newton'sche Potential*. B. G. Teubner, 1877.

[OA14] Hamed Orojloo and Mohammad Abdollahi Azgomi. A method for modeling and evaluation of the security of cyber-physical systems. In *2014 11th International ISC Conference on Information Security and Cryptology*, pages 131–136, 2014.

[PPH13] Mahendra Piraveenan, Mikhail Prokopenko, and Liaquat Hossain. Percolation Centrality: Quantifying Graph-Theoretic Impact of Nodes during Percolation in Networks. *PLOS ONE*, 8(1):1–14, 01 2013.

[PRT10] R. Pietrantuono, S. Russo, and K. S. Trivedi. Online Monitoring of Software System Reliability. In *2010 European Dependable Computing Conference*, pages 209–218, 2010.

[RC18] Gregory Roth and Hal Caswell. Occupancy time in sets of states for demographic models. *Theoretical Population Biology*, 120:62–77, 2018.

[SJT05] Vibhu Saujanya Sharma, Pankaj Jalote, and Kishor S. Trivedi. Evaluating Performance Attributes of Layered Software Architecture. In George T. Heineman, Ivica Crnkovic, Heinz W. Schmidt, Judith A. Stafford, Clemens Szyperski, and Kurt Wallnau, editors, *Component-Based Software Engineering*, pages 66–81, Berlin, Heidelberg, 2005. Springer Berlin Heidelberg.

[Spr18] Jeremy Sproston. *Qualitative Reachability for Open Interval Markov Chains*, page 146–160. Springer International Publishing, 2018.

[ST06] Vibhu Saujanya Sharma and Kishor S. Trivedi. Reliability and Performance of Component Based Software Systems with Restarts, Retries, Reboots and Repairs. In *2006 17th International Symposium on Software Reliability Engineering*, pages 299–310, 2006.

[SVA06] Koushik Sen, Mahesh Viswanathan, and Gul Agha. Model-Checking Markov Chains in the Presence of Uncertainties. In Holger Hermanns and Jens Palsberg, editors, *Tools and Algorithms for the Construction and*

Analysis of Systems, pages 394–410, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.

[Wal96] Peter Walley. Measures of uncertainty in expert systems. *Artificial Intelligence*, 83(1):1–58, 1996.